
Signal and Noise: A Framework for Reducing Uncertainty in Language Model Evaluation¹

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Abstract⁴

Developing large language models is expensive and involves making decisions with small experiments, typically by evaluating on large, multi-task evaluation suites. In this work, we analyze specific properties which make a benchmark more reliable for such decisions, and interventions to design higher-quality evaluation benchmarks. We introduce two key metrics that show differences in current benchmarks: signal, a benchmark’s ability to separate better models from worse models, and noise, a benchmark’s sensitivity to random variability between training steps. We demonstrate that benchmarks with a better signal-to-noise ratio are more reliable when making decisions at small scale, and those with less noise have lower scaling law prediction error. These results suggest that improving signal or noise will lead to more useful benchmarks, so we introduce three interventions designed to directly affect signal or noise. For example, we propose that switching to a metric that has better signal and noise (e.g., perplexity rather than accuracy) leads to better reliability and improved scaling law error. We also find that filtering noisy subtasks, to improve an aggregate signal-to-noise ratio, leads to more reliable multi-task evaluations. We also find that averaging the output of a model’s intermediate checkpoints to reduce noise leads to consistent improvements. We conclude by recommending that those creating new benchmarks, or selecting which existing benchmarks to use, aim for high signal and low noise. We use 30 benchmarks for these experiments, and 375 open-weight language models from 60M to 32B parameters, resulting in a new, publicly available dataset of 900K evaluation benchmark results, totaling 200M instances.

 allenai/signal-and-noise  datasets/allenai/signal-and-noise⁷

1 Introduction⁸

Language model development is expensive. During the development process, researchers need to make decisions such as what architecture to use, what training methods to employ, and what data to train on. These decisions rely on measuring phenomena at smaller, more economical scales, then hoping the trends measured hold for large scale models. This paradigm exists across the research community; many papers experiment with small baselines then scale up the best-performing model [31, 17, 38, *inter alia*], and there has been extensive research on using scaling laws to predict the performance of larger models [9, 19, *inter alia*]. While there is a large and ever-growing number of benchmarks, prior work has shown these scaling procedures only works for some benchmarks and not others [66, 56, 15, 50]. This poses a significant challenge because, as we develop more general-purpose language models, developers need to be evaluating on even more diverse benchmarks, some of which may not be well-suited for this critical approach. We need a deeper understanding

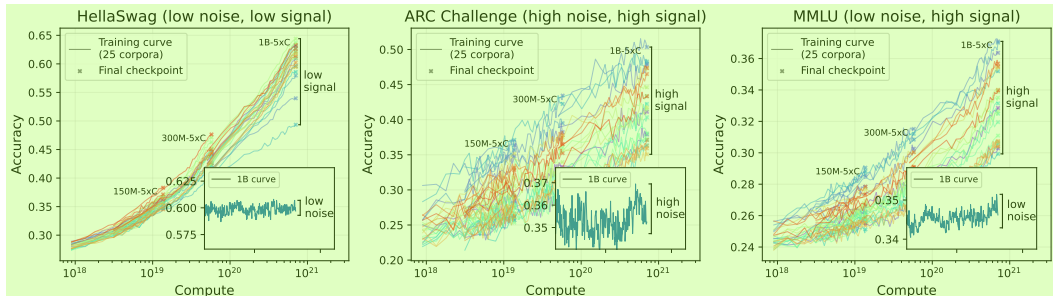


Figure 1: Training curves for the 25 pretraining corpora in DataDecide [38] on three development benchmarks across different model sizes – the ordering of different model pre-training corpora, shown by different colors, at a small scale (e.g., 150M) should agree with ordering at a larger scale (1B), implying better decision accuracy. We hypothesize that one indicator of decision accuracy is the ratio between the signal (main plot) and the noise of scores within a single training run (inset axis). In this work, we quantify the signal-to-noise ratio at different compute scales, and in later sections, show that it is predictive of large scale phenomena like decision-making error.

of what intrinsic properties we can measure to tell if a benchmark provides useful information, if it needs to be reformulated, or if it is best discarded altogether.

To formalize this setup, we study two common experimental settings for language model development: (i) train a pair of small models (e.g., on different pretraining corpora) and use their ranking to predict the ranking of two large models [38], and (ii) fit a scaling law on a set of small models and predict the performance of a large model [19, 3]. We hypothesize that the ability to predict both settings are related to a measure which is cheaper to compute and easier to improve: signal and noise. Signal measures how spread out scores are for different models on a single benchmark and noise measures the variability of a benchmark score during training.

To illustrate the connection from signal and noise to an experimental setting, consider an example of comparing models trained using different pretraining corpora (illustrated in Figure 1); the tasks where scores are either too close (HellaSwag, left) or too noisy (ARC Challenge, center) are the benchmarks where we would be less confident that a ranking of models at a small scale would hold at a large scale. Following this observation, we show in Section 4 that the signal-to-noise ratio (SNR) is highly correlated with the likelihood that a ranking of models at a small scale will hold at a large scale, and then show that noise is highly correlated with the prediction error of a scaling law fit.

Based on these observations, in Section 5 we propose a set of interventions designed to reduce noise or increase signal, and then we measure their impact on our experimental setups of decision accuracy and scaling law error. For example, we show that by averaging out the checkpoint-to-checkpoint noise for a model, we improve our ability to predict performance of large models from small models. We also show that it is possible to find subsets of existing benchmarks that have higher signal-to-noise ratios than the full evaluation sets, and that even though those subsets can have fewer than half as many instances, they improve both experimental setups. Finally, we show that SNR can be used to improve metric construction, where choosing a metric that has better SNR leads to consistent improvements on a wide variety of benchmarks.

Our core contributions are as follows: (i) we introduce definitions for signal, noise, and signal-to-noise ratio in the setting of evaluating language models, and show this framework is useful for measuring the utility of benchmarks, and (ii) we demonstrate interventions based on this framework which improve both prediction settings. Our core results evaluate 465 language models on 30 benchmarks across 14 model sizes. We release our data, evaluation results, and trained models.

2 Predicting Large Model Phenomena with Small Models

Using small scale experiments to make predictions about large model behavior is ubiquitous in language model development [27, 60, 31, 42]. This process can take many forms. For example, finding a good mix of data from multiple sources to train on typically involves evaluation of small models to calculate an optimal weighting of datasets, then training a large model on the optimized

mix [35, 66]. In Blakeney et al. [5], mid-training runs on a sample of candidate pretraining datasets are used to estimate the quality of training from-scratch. Dubey et al. [17] predicted the downstream task using scaling laws to compare candidate data mixes. Hyperparameter transfer methods, such as maximal update parametrization (μP), also rely on small scale experiments [68]. However, the results from small scale experiments are not always reliable. Work on so-called emergent capabilities [65] shows that for some benchmarks, language model performance only rises above random chance for models trained at large compute budgets. Later work has further explored emergence behavior in particular tasks, such as MCQA tasks [67] or generative math and code tasks [57], or by observing the capabilities of open-weight models [51].

While these different experimental setups are all important, we focus on two straightforward and common setups in making data decisions for language model development: decision accuracy and scaling law prediction error. In this section, we present the motivation for both experimental settings, and in Section 3 we show how the signal-to-noise ratio is an effective framework for predicting how useful a benchmark in these scenarios.

2.1 Decision Accuracy and Scaling Law Prediction Error

Decision Accuracy. Consider a scenario where a practitioner intends to train a large model, and needs to decide between training on Dataset a or Dataset b to get the best performance on some downstream task, represented by a scalar $B(\cdot)$. A simple and intuitive approach is to train a small model s_a on Dataset a and another, s_b , on Dataset b , then choose the dataset that led to the best downstream task performance for training the large model. We evaluate this procedure by training two large models, m_a and m_b , one on each of the datasets, and see if the ranking of the two small models, s_a and s_b , on the benchmark is the same as for the large models.¹ In the scenario where we are deciding between more than two choices, we consider pairwise rankings between all pairs $\mathcal{P}$. Following Magnusson et al. [38] we refer to this small-to-large agreement as “decision accuracy”:

$$\text{Decision Accuracy} = \frac{1}{|\mathcal{P}|} \sum_{(a,b) \in \mathcal{P}} \mathbb{I}[\text{sign}(B(s_a) - B(s_b)) = \text{sign}(B(m_a) - B(m_b))] \quad (1)$$

We use models of 7 sizes (from 60M parameters up to 1B parameters) trained on 25 different pretraining corpora from Magnusson et al. [38]. Our prediction task is to use a set of small models (e.g., 60M parameter models) to predict the ranking of the 1B models on a given benchmark (e.g., MMLU). High decision accuracy means the ranking of the small models accurately predicts the ranking of the large models on that benchmark; this is an indication that the benchmark is useful for this process of using small models to make decisions about which dataset to train on. We illustrate an example of this in Figure 1, which shows training curves for 25 data recipes on 3 model sizes. We hypothesize that if model scores are very close together, or the evaluations are very noisy, it is more likely that the ranking from small to large models will change, leading to worse decision accuracy; we formalize and test this hypothesis in the following sections.²

Scaling Law Prediction Error. Scaling laws [27, 24, inter alia] have been used extensively to predict the validation loss of a large model using a set of smaller “scaling law” models. Recent work has also used scaling laws to predict downstream task performance [19, 3] by first predicting task loss then using the predicted loss to predict task performance (e.g., accuracy); this is the setup we use in this work. The prediction error for the scaling law fit is defined as the relative error between the predicted and true performance of the large model: $\text{Prediction Error} = \frac{|\text{Measured Value} - \text{True Value}|}{|\text{True Value}|}$.

Calculating prediction error requires training a set of scaling law models on the same corpus with varying tokens/sizes (e.g., 190M to 1B params), training a large model (e.g., 13B), and fitting a scaling law to the smaller models to predict the larger model performance.³ We describe the scaling law functional form and fitting details in App. A.1, following the setup in Bhagia et al. [3].

¹Training multiple large models is too expensive for most development scenarios, but is necessary to evaluate how accurate this process is.

²We observe similar findings on other rank agreement metrics, like Spearman rank correlation (Table 3). Decision accuracy, in particular, is equivalent to Kendall’s tau modulo a scale and shift (App. A.2).

³Scaling law predictions can be used to make development decisions (e.g., about which training dataset is best) by training a set of models and fitting a scaling law for each option being considered [17], but in this work we just evaluate scaling law error directly.

2.2 Evaluation Dataset 1

We perform our analysis using existing development benchmarks and models: 2

Models. Our set of models includes: (i) a suite of scaling law models from 190M to 3.2B, with a 3 corresponding target at 7B and 13B [3], (ii) a suite of 25 models each trained with different pre-training corpora from 60M to 1.3B [38], (iii) the final 30 checkpoints for OLMo 2 1B, 7B, 13B and 32B [42], and (iv) 73 open-weight base models. Additionally, in our comparison between sources of modeling noise in §3.1, we train and release 20 1B models, with 10 models trained varying the data order initialization and 10 varying the random seed initialization, along with evaluation on 3.2K intermediate checkpoints.

Benchmarks. We evaluate 30 development tasks which we categorize as knowledge QA, math, and 4 code. We use the OLMES [22] standard where applicable, and reproduce the OLMo 2 evaluation setup [42] for all other benchmarks. Following Gadre et al. [19], we also include multi-task averages for each group, and for the OLMES core tasks. For our test of subset selection in §5.1, we include a synthetically generated benchmark, generated using AutoBench [32].

We include full details on the sets of models and benchmarks in App. A.5. 5

3 Quantifying Signal and Noise 6

To illustrate the impact of noise on a decision-making setup, Figure 1 shows training curves for 25 7 1B models trained with different data recipes and, in inset plots, the training curve for a single 1B model on three tasks. Some tasks (left, HellaSwag) exhibit low noise between training checkpoints but low signal between models, and others (center, ARC-Challenge) exhibit high noise and high signal. In this section we define signal and noise, and define two simple metrics to estimate the signal-to-noise ratio that can be calculated from a set of model evaluations on a given benchmark.

3.1 Measuring Noise 8

There are numerous sources of noise in the language model development pipeline. Previous work 9 has shown multiple training runs under the same configuration can lead to different performance as a result of a different initialization or data order [14, 13]. In addition, as illustrated in Figure 1, performance can even vary significantly from one checkpoint to the next: within the final 30 checkpoints of training for 1B models on ARC Challenge, we observe a range of 1.7% accuracy. With these motivations, we consider four potential noise measurements, each calculated on using evaluation on a single benchmark: (i) training multiple models and varying only the random initialization, (ii) training models and varying the training data order, (iii) measuring the total checkpoint-to-checkpoint noise across a full, single training run, and (iv) measuring the checkpoint-to-checkpoint noise of the final n checkpoints of a single training run. We formalize these definitions in App. A.3.

To get estimates for four potential sources of noise, we train 10 different 1B-5xC models varying the 10 initialization and data orders, and evaluate all intermediate checkpoints. We find that the initialization noise, data order noise, and checkpoint-to-checkpoint noise across the whole training run all correlate highly with the relative standard deviation of the final n checkpoints (R^2 of 0.82, 0.86, and 0.95, respectively, see Figure 7; and see the training curves in Figure 19). These results lead us to define noise as the relative standard deviation of the final n checkpoints, as this requires no additional training cost and only uses the final n checkpoints rather than the full training curve. We define noise as: $\text{Rel. Std.}(m) = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (m_i - \bar{m})^2} / \bar{m}$.

3.2 Measuring Signal 11

A benchmark is most useful during language model development if it can detect a true difference 12 between a good model and a poor model, assuming a true difference exists between the models in the ability that the benchmark aims to measure. This statistical power is what enables us to use small models for development decisions like training dataset to use. To formalize this idea, we consider a benchmark to have high signal when models evaluated on it have a wide and evenly distributed range of scores. We measure signal using a metric from the numerical integration literature: dispersion, calculated as the maximum difference between the scores of any two models, divided by the mean

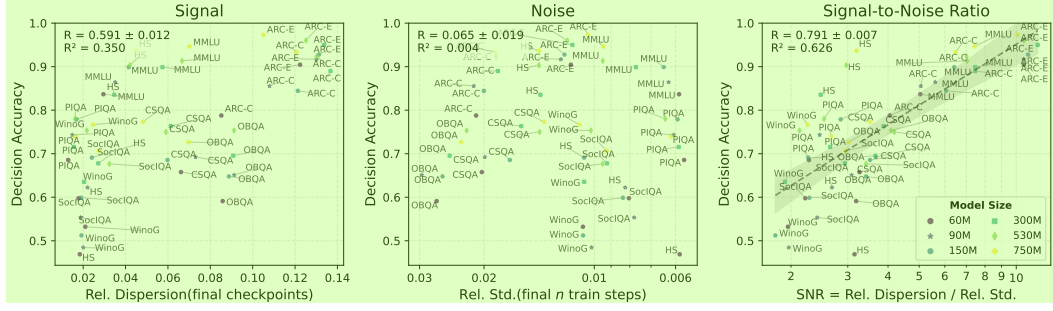


Figure 2: Signal, noise, and signal-to-noise ratio (x -axis) vs. decision accuracy (y -axis), (see Section 2 for definitions). The signal alone (left) and noise alone (center) have low correlation with decision accuracy, while the signal-to-noise ratio (right) is correlated with decision accuracy. The signal-to-noise ratio gives us information about whether a benchmark is useful during development, as high decision accuracy (and signal-to-noise ratio) means development decisions made at a small scale generalize to large scale models.

score of all models to account for different scales. This metric is designed specifically to measure how well a set of points cover a space; that is, how spread out the points are from each other. We also considered 20 different measures of spread, including variance, mean pairwise distance, Gini coefficient, etc., in Appendix A.4.

In the following section we introduce signal-to-noise ratio, and find that this definition of signal leads to signal-to-noise ratio with the highest correlation with decision accuracy. We define signal as $\text{Rel. Dispersion}(M) = \max_{j,k} |m_j - m_k| / \bar{m}$, the normalized maximum difference between any pair of models j, k .

3.3 Measuring Signal-to-noise Ratio

Using our measures of signal (§3.2) and noise (§3.1), we propose measuring the signal-to-noise ratio. For both measures, we first divide by the average to be independent of particular units (e.g., to compare accuracy to unbounded task perplexity). We define the signal-to-noise ratio:

$$\text{Signal-to-Noise Ratio} = \frac{\text{Rel. Dispersion}(\text{final train checkpoint})}{\text{Rel. Std.}(\text{final } n \text{ train checkpoints})} \quad (2)$$

where signal (Rel. Dispersion) is measured over a population of models trained using a similar compute budget, and noise (Rel. Std.) is measured over the final n intermediate training checkpoints of a single model. We emphasize that, while this is one particular instantiation of the signal-to-noise ratio, our framework is designed to be independent of a particular metric: we find many other measures of signal produce similar results in Appendix A.4 and measures of noise have high correlation in Appendix A.3.

4 Signal and Noise Correlate with Better Predictions

In this section, we show that the signal-to-noise ratio correlates with decision accuracy for small scale experiments, and that the noise of the target model correlates with scaling law prediction. These findings motivate our use of SNR to improve benchmarks' statistical properties in Section 5.

4.1 Higher signal-to-noise ratio indicates higher decision accuracy

Setup. We hypothesize that a higher signal-to-noise ratio makes it easier to distinguish between models. To test this, we measure decision accuracy using the ranking of the small DataDecide models (60M to 750M) to predict the ranking of the large DataDecide model (1B). To calculate signal we use the final checkpoint of each of the 25 small models, and to calculate noise, we use the standard deviation around the final 5 checkpoints of the small-scale models. Since we have a measure of noise for *each* model, we use the average of the noise across the small models.

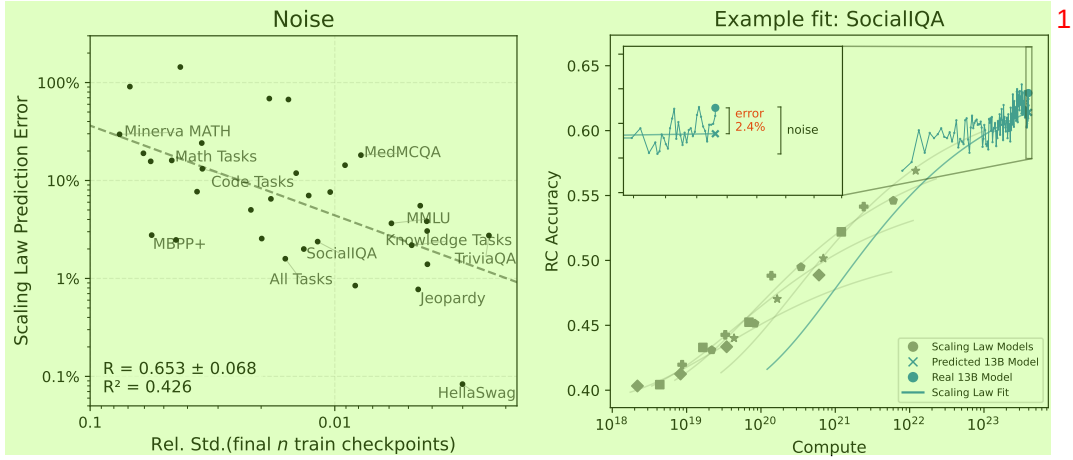


Figure 3: **Left:** Correlation between the noise and scaling law prediction error (see Section 2 for definitions). We observe benchmarks with a lower noise around the scaling law target (x -axis) also exhibit lower error (y -axis). **Right:** Example of scaling law for one benchmark (SocialIQA), with examples on all benchmarks in Figure 15. We conjecture that the noise of the target model (see inset axis) acts as a bound on the true minimum scaling law error; if the observed scaling law error below this noise, then the error is only possible by random chance. Therefore, when benchmarks exhibit a similar scaling law error but different noise (e.g., MBPP+, SocialIQA and TriviaQA; see Figure 15), we argue that those with the lowest noise are better.

Signal-to-noise is predictive of decision accuracy. Figure 2 shows the signal, noise and signal-to-noise ratio plotted against the decision accuracy across the OLMES benchmarks. While the signal or noise alone do not correlate with decision accuracy, we find a strong correlation between SNR and decision accuracy ($R = 0.791$, $R^2 = 0.626$). We conclude that benchmarks which have higher SNR at small scales exhibit higher decision accuracy, and are more likely for their results to hold at a larger scale. In Appendix B.1, we observe benchmarks with a higher SNR also exhibit lower variance when calculating decision accuracy using different checkpoints around the end of training.

4.2 Tasks with higher noise also have higher scaling law error⁴

Setup. We fit scaling laws to predict the performance of OLMo 2 13B using final checkpoint of the set of scaling models trained by Bhagia et al. [3]. We calculate the scaling law prediction error as the relative error of the predicted and final 13B checkpoint. To estimate the noise, we calculate the relative standard deviation of the final 30 checkpoints of the 13B training run, each spaced 1000 training steps until the end of training.⁴ We hypothesize that the range of the final k checkpoints of the prediction target (the large, 13B model) acts as an lower-bound on the true minimum scaling law prediction error. An example of the prediction error and noise around the prediction target is illustrated using SocialIQA in Figure 3 (right). Assuming a scaling law with no bias, we expect tasks with a lower standard deviation of the prediction target to also have a lower prediction error.

Noise measures the reliability of scaling law prediction errors. In Figure 3 (left), we show the scaling law error and standard deviation for predicting the 13B model performance over 30 tasks. We observe a correlation between the standard deviation of the prediction target and the prediction error across tasks ($R = 0.653$, $R^2 = 0.426$), however the fit is not perfect. For example, we observe four tasks (MBPP+, SocialIQA, MMLU and TriviaQA) which exhibit similar error (around 2–3%), but exhibit different amounts of noise around the prediction target. For these benchmarks with similar error but lower noise, we can be confident that the error we observe from the single scaling law fit is the result of the true error of the scaling law fit rather than random chance. In practice, we recommend practitioners prefer making decisions based on scaling law predictions using tasks with low error *and* low noise.

Previous work has fit multi-task averages to predict scaling laws. In particular, Gadre et al. [19] find that the error from the individual tasks in their work to be too difficult to predict accurately. In Figure

⁴We found 30 checkpoints to be an adequate trade-off between sample size and compute cost. We provide guidance on selecting n when calculating noise, and its impact on experimental results, in Appendix A.3.2.

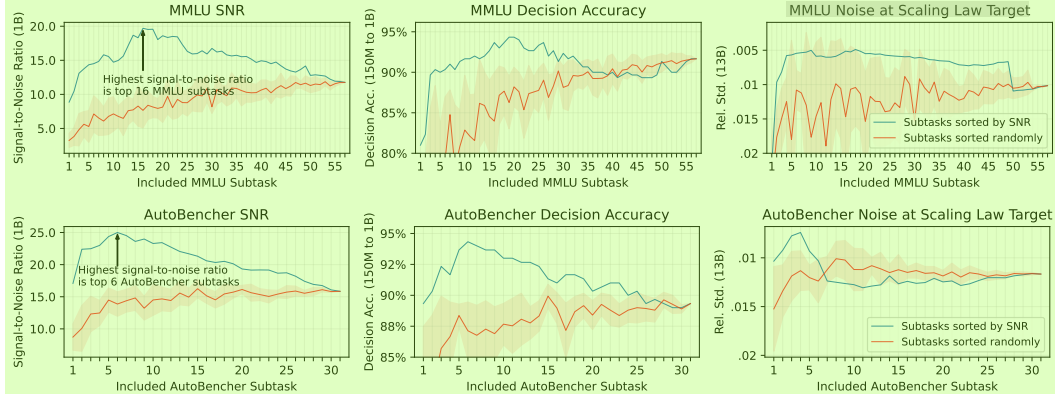


Figure 4: Evaluating an intervention designed to increase signal-to-noise ratio (SNR): selecting subsets of a benchmark (Top: MMLU; Bottom: AutoBench) that have higher SNR dramatically improves decision accuracy and the noise of the scaling law prediction target. MMLU and AutoBench are made of different subtasks; for each benchmark we sort its subtasks by their SNR, then greedily add subtasks to our subset in order of decreasing SNR (left to right). Despite the subsets made in this way having fewer test instances, we find subsets of MMLU (e.g., with 16 subtasks) and of AutoBench (e.g., with 6 subtasks) that have higher SNR than the full sets, and also have better decision accuracy and noise around the scaling law target. Named subtasks in Figure 16 in Appendix.

3 we also plot results for multi-task averages for each task group (‘Knowledge’, ‘Math’, ‘Code’) and an average across ‘All Tasks’. We find that some individual tasks are easier to predict than multi-task averages, and have lower noise around the prediction target. In particular, generative tasks like TriviaQA or Jeopardy which evaluate the exact match of a short-form generation exhibit lower error than the multi-task averages, and exhibit lower noise around the prediction target. For practitioners, we argue using individual tasks may be a better decision in some cases than the multi-task average, if that task better represents the ability than a multi-task average.

Our core results report SNR at the scales of our experimental settings for decision accuracy and prediction error. However, SNR can be calculated at any model size, so we show how the signal-to-noise ratio changes for tasks at larger 1B, 7B, 13B and 32B scales in Appendix B.3.

5 Improving Predictions by Improving SNR 5

In this section, we introduce three interventions designed to improve the signal, noise, or SNR: 6 filtering subtasks by SNR (§5.1), averaging checkpoint scores during a training run (§5.2), measuring language modeling loss over the test set using bits-per-byte (§5.3). In each setup, we show using signal-to-noise ratio to intervene on the task improved the resulting error in both prediction settings.

5.1 Filtering noisy sub-tasks improves signal-to-noise ratio 7

Setup. Many tasks are a macro-average of subtasks. We hypothesize that some subset of subtasks is 8 usually higher quality than the rest of the set, and that the signal-to-noise ratio may be an indicator of high quality subtasks. To test this, we first calculate the signal-to-noise ratio of each subtask, then rank the subtasks by signal-to-noise ratio and greedily add the highest SNR subtasks. As a baseline, we randomly shuffle the subtasks, and report the average of 10 calculations of each metric, with the shading indicating ± 1 standard deviation.

Results. We show results in Figure 4. For MMLU, using only 16 subtasks had a higher signal-to- 9 noise ratio than using the full test set. For AutoBench, we observe the same but with only 6 tasks. The lower signal-to-noise ratio also led to a higher decision accuracy: +2.6% for MMLU and +5% for AutoBench by using the high SNR subset compared to the full benchmark. We hypothesize that the quality of a task subset may influence that task’s signal-to-noise ratio. To test this, we use the data collected from MMLU Redux, which identified MMLU subtasks with high labeling error [21]. We find that out of the 20 MMLU subtasks which contain errors in least 5% of instances, half of these subtasks (10 of 20) are also in the lowest 20 tasks sorted by their signal-to-noise ratio. This presents

Table 1: Evaluating an intervention designed to average out noise: for a given model on one benchmark, we calculate its score as the average of the scores of its final k checkpoints (evaluated using bits-per-byte task formulation). **Left:** On small models used to make predictions (‘Avg. Pred.’), or to the large target models (‘Avg. Target’), or both (‘Avg. Both’), decision accuracy improves. * indicates the decision accuracy is the same across columns. **Right:** On small models used to fit scaling laws (‘Avg. Train’), scaling law error improves. We show results on a subset of benchmarks, and report all benchmarks and the primary metric (accuracy, exact match, pass@1) in Tables 5 and 6.

Decision Accuracy (60M-5xC to 1B-5xC), %					Prediction Error (13B-5T), Abs. %		
Task ↓	Final Ckpt	Avg. Pred.	Avg. Target	Avg. Both	Task ↓	Final Ckpt	Avg. Train
Knowledge QA Tasks					Knowledge QA Tasks		
ARC Challenge	94.5	94.9	94.3	94.6	HellaSwag	0.31	0.16
HellaSwag	92.4	93.1	93.1	94.0	CommonsenseQA	0.59	0.46
ARC Easy	92.1	92.2	91.9	92.0	Jeopardy	0.57	0.54
MMLU	91.5	91.6	91.6	91.6	SocialIQA	0.50	0.59
AutoBench	88.5	88.9	89.1	89.6	PIQA	0.89	1.01
MMLU Pro	90.0	89.4	90.0	89.3	MMLU	1.68	1.74
AGI Eval	86.3	86.7	86.5	87.0	MMLU Pro	1.76	1.75
MedMCQA*	86.6	86.6	86.6	86.6	AGI Eval	1.89	1.98
Jeopardy	84.4	84.4	84.8	85.0	BoolQ	4.13	2.48
TriviaQA	83.5	84.3	83.8	84.6	TriviaQA	2.33	2.62
OpenBookQA	81.4	81.7	81.6	82.0	SQuAD	2.80	2.79
SocialIQA	79.9	79.5	79.4	79.0	OpenBookQA	4.02	3.38
PIQA	72.5	72.9	71.9	72.0	AutoBench	3.86	3.69
CommonsenseQA	65.8	66.2	65.4	65.6	ARC Easy	5.13	5.13
BoolQ	63.7	64.2	63.5	64.0	MedMCQA	7.72	7.98
SQuAD	60.8	60.4	62.0	61.6	ARC Challenge	8.44	8.43
Knowledge 19-Task Avg.	71.3	71.5	71.7	71.7	Knowledge 19-Task Avg.	1.43	1.20
Code Tasks					Code Tasks		
HumanEval*	95.6	95.6	95.6	95.6	MBPP	2.57	1.79
MBPP*	95.3	95.3	95.3	95.3	HumanEval	7.71	8.85
Code 4-Task Avg.*	96.7	96.7	96.7	96.7	Code 4-Task Avg.	3.15	2.75
Math Tasks					Math Tasks		
Minerva MATH*	90.0	90.0	90.0	90.0	Minerva MATH	1.08	0.98
GSM8K*	76.6	76.6	76.6	76.6	GSM8K	7.46	3.85
Math 6-Task Avg.*	88.3	88.3	88.3	88.3	Math 6-Task Avg.	11.33	2.30
All 30-Task Avg.	68.9	70.7	69.5	71.3	All 30-Task Avg.	1.03	0.86

evidence that low SNR may indicate low quality tasks, and we believe this is a good opportunity for future work in evaluation development.

Intuitively, a benchmark developer may increase the statistical power of a comparison between models: by sampling more data by the original process used to construct the benchmark, in order to make a benchmark larger [64], or collect a larger number of tasks in an evaluation suite [58]. Our evidence in Figure 4 suggests that larger benchmarks may not necessarily be better for comparing models. We further explore this phenomenon in App. B.2 by sub-sampling instances of benchmarks, finding some benchmarks can exhibit a higher SNR despite having 10 times fewer instances.

5.2 Averaging checkpoint-to-checkpoint noise leads to better predictions

Setup. Typically, models are only compared using the evaluation of the final checkpoint. In the previous sections, we argued that noise is a good indicator of whether we can use a benchmark to predict a large scale phenomenon. In this section, we want to measure the effect of averaging this particular source of step-to-step noise, as a way of improving our ability to make a prediction. In the decision accuracy setting, we can average the results of the small model, the large model (in this case, the 1B model), or both. In the prediction error setting, averaging the small models will help in fitting the scaling law, but averaging the target model will just make the result more reliable, so we average the target model in both settings and only change whether we average the models used to fit the scaling law. Finally, we introduce an additional way to average step-to-step noise during a training run, by evaluating whether the ranking of the 1B models during training agrees with the ranking at the end of training. Note, as our measure of noise is between intermediate training checkpoints, we are only reducing one of many sources of modeling noise.

Results on Final Checkpoints. In Table 1, we observe averaging the noise improved both measures of error. Averaging noise improved decision accuracy by +2.4% for the 30-task average, this procedure improved decision accuracy in all but two tasks. For reducing the scaling law prediction error, averaging the training checkpoints improved prediction error for 20 of 30 tasks.

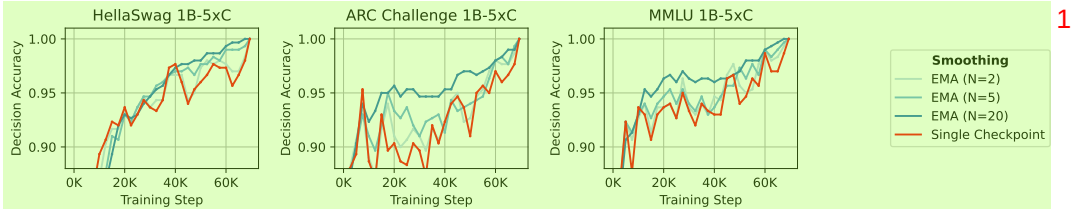


Figure 5: When stopping a training run early, averaging the checkpoint-to-checkpoint noise improves the decision accuracy between an intermediate and the final training step. Shown are decision accuracy from early-stopping for HellaSwag, ARC-C and MMLU by using both a single checkpoint and the exponential moving average (EMA), with all tasks included in Figure 18.

Experiment Setting → Metric →	SNR (↑)		Rel. Error (↓), %		Decision Acc (↑), %	
	Primary	BPB	Primary	BPB	Primary	BPB
Knowledge QA Tasks						
TriviaQA	27.9	61.8	2.5	0.5	68.3	85.3
SQuAD	23.8	29.0	7.6	27.8	59.7	61.7
ARC Easy	21.0	64.6	5.3	0.8	93.0	93.0
Jeopardy	20.2	22.6	3.5	18.6	82.0	83.0
AutoBench	15.9	31.3	0.2	4.5	89.3	89.3
HellaSwag	11.8	14.9	1.4	1.0	74.3	95.3
MMLU	9.8	35.9	4.3	0.4	89.0	92.0
ARC Challenge	6.6	44.8	9.7	2.1	83.3	95.0
SocialQA	5.5	48.0	0.4	1.9	55.0	80.0
PIQA	4.2	8.8	0.5	1.3	73.3	72.7
AGI Eval	2.5	19.5	13.7	3.4	58.7	88.0
Knowledge 19-Task Avg.	13.7	44.3	0.8	1.0	79.0	80.0
Math Tasks						
Minerva MATH	1.9	88.6	11.9	1.9	51.0	90.0
GSM8K	1.2	7.0	38.6	5.9	46.0	76.7
Math 6-Task Avg.	1.8	22.6	46.0	5.0	42.3	88.3
Code Tasks						
HumanEval	6.1	25.1	9.2	7.9	74.3	95.7
MBPP	2.0	41.8	23.6	1.0	68.3	95.3
Code 4-Task Avg.	5.5	42.0	29.5	9.7	80.3	96.7
All 30-Task Avg.	10.0	31.5	2.3	0.4	77.0	83.7

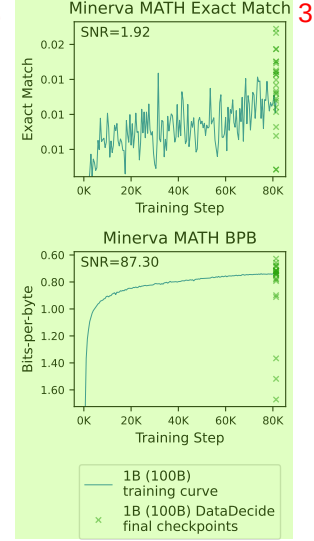


Figure 6: Impact of changing benchmark metric to bits-per-byte (BPB) from the primary score (e.g., accuracy, pass@1, etc.). **Left.** Columns are (i) SNR of 1B models trained to 100B tokens; (ii) scaling law prediction error of 1B (and smaller) models used to predict 13B model performance; (iii) decision accuracy for using 150M model to predict 1B model ranking. For almost all tasks at the scales explored here, bits-per-byte shows a higher SNR, and lower scaling law prediction error, and higher decision accuracy than the primary score. Full results across 30 benchmarks and model scales in Table 17. **Right.** Example of primary metric and BPB on a single 1B (100B tokens) training curve (blue curve) and the final checkpoint of 25 models for Minerva MATH (green 'x's). Visually, the BPB training curve is smoother, corresponding to a higher SNR and a lower error in the prediction settings reported in the table, with all tasks in Figure 14.

Results on Early Stopping. Another prediction setting is to determine whether the ranking of two partially trained models will exhibit the same order at the end of training. We hypothesize that averaging the step-to-step noise will similarly improve this setting. In Figure 5, we report the decision accuracy for early stopping by using a single checkpoint (red), compared to an exponential moving average of the training curve (blue). We find for almost any training step, applying smoothing led to a higher decision accuracy when comparing models during training. In both settings, reducing the checkpoint-to-checkpoint noise allowed a more accurate extrapolation.

5.3 Measuring bits-per-byte improves benchmark signal-to-noise ratio

Setup. Recent work has begun to evaluate by using the test set as a perplexity set, with the intuition that the discontinuous metrics like accuracy or exact match erode the relationship between the language modeling perplexity and the downstream metric [54, 25]. We aim to measure whether the intervention to use a continuous metric improves the signal-to-noise ratio and corresponding error. We calculate the bits-per-byte (BPB) using the correct continuations of each test set – the bits-per-byte is the negative log likelihood of the correct answer divided by the number of UTF-8

bytes in the answer string [20, 37]. We compare BPB to the ‘primary’ task metric (accuracy, exact match, pass@1, etc.) on the signal-to-noise ratio, and whether it improves decision-making using decision accuracy from 150M to 1B and reduces the scaling law prediction error at 13B.

Results. In Figure 6 we report the signal-to-noise ratio, scaling law error and decision accuracy for benchmarks using BPB instead of the primary metric, along with an example training curves for Minerva. Most benchmarks have higher signal-to-noise ratio when using the BPB, particularly generative math and code benchmarks like GSM8K (1.2 to 7.0) and MBPP (2.0 to 41.8). To verify this improvement in signal-to-noise ratio corresponds to an improvement in our decision-making setups, we observe an improvement in decision accuracy at the small scale for 90.0% of all benchmarks and a lower scaling law prediction error for 73.3% of all benchmarks. We see BPB results in dramatic improvement for tasks that small scale models are not able to accomplish at all, primarily generative tasks. Our results confirm that BPB is a useful metric is both a higher quality development benchmark, particularly for challenging tasks at small scales that do not show above random-chance signal.

6 Related Work and Discussion

Predicting model behavior at large scales is crucial aspect to language model development, as discussed in the beginning of §2. Noise within evaluation benchmarks is frequently studied as the intrinsic noise of the dataset [2, 7, 40, 6], rather than the noise as a result of differences in the model during training. Closest to our work is Madaan et al. [36], which report a measure of SNR using the benchmark score of a *single* model and noise using 10 seed models, rather than a population of models. We find that the noise of a single model alone, while a useful measure of modeling noise, is not sufficient as a measure of correlation to decision accuracy (§4), and show the step-to-step noise is a cheap alternative to seed noise. Similarly, Kydlíček et al. [29] focus on identifying high quality translations of tasks, but do not focus on decision making. Finally, EvalArena [63] also reports a measure of SNR using the final checkpoints of a small/large model pair (e.g., Llama 3 7B vs. 70B). While statistical measures based on intrinsic noise rather than modeling noise are important indicators of dataset noise, we find that many benchmarks may have low statistical variability but high checkpoint-to-checkpoint noise (such as BoolQ, as observed in Figure 9), which can only be captured with a measure of modeling noise.

Interventions to improve evaluation have been well explored, such as constructing higher quality benchmarks by identifying errors [62, 21], expanding test sets [64], selecting high quality instances from benchmarks [45], or generating entirely new synthetic benchmarks from a model [32]. These works typically justify their decisions using inter-annotator agreement, or a high correlation with the original benchmark. We believe this body of work can benefit from verifying their methods using SNR, rather than noise or reconstruction error alone, to indicate whether the benchmark serves as a useful development tool.

Notably, this scope of our connection between the signal-to-noise ratio and predicting large scale phenomena is limited to the two decision accuracy and prediction error settings, and only studies the noise of the model during training. Future work may explore how signal-to-noise ratio indicates other small-to-large phenomena [65, 57], and the effects of additional sources of noise on the ability to extrapolate from small-scale experiments, such as from the evaluation configuration [55, 22].

In this work, we identify signal and noise as a cheap way of estimating whether a benchmark is useful in predicting large-scale phenomena with small scale experiments. We conclude that new benchmark development should use these measures of modeling noise as a guide for building evaluation tools for model developers, and practitioners adopt interventions, such as those introduced in this work, that improve their ability to compare models.

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A Methodology Details¹

A.1 Scaling Law Details²

Hoffmann et al. [24] models the improvement for larger model training budgets as a power function,³ proportional to the model parameters N and training tokens D , with the exact functional form and prediction setup varying between work [44]. Recent work has begun using the downstream task as the prediction target [17, 19], in this work we follow Bhagia et al. [3] by fitting a scaling law function to the language modeling loss over the correct continuation, then from the task loss to the downstream evaluation. We use the following functional form:

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E, \quad U(L) = \frac{a}{1 + e^{-k(L-L_0)}} + b \quad (3)$$

We follow the same methodology as Bhagia et al. [3] and use the Huber loss to fit $L(N, D)$ and use a⁵ non-linear least squares optimizer to fit $U(L)$. The prediction error is defined as the relative error of the scaling law fit:

$$\text{Prediction Error} = \frac{|\text{Measured Value} - \text{True Value}|}{|\text{True Value}|} \quad (4)$$

A.2 Decision Accuracy Details⁷

Decision accuracy is one of many rank agreement metrics we could use to show that models trained⁸ across pre-training corpora agree at a small scale and a large scale. We present two alternatives here:

Kendall’s τ . Here, rather than report Kendall’s τ , we show it is proportional to decision accuracy.⁹ Kendall’s τ is defined as the difference between the concordant pairs C and discordant pairs D , divided by the total pairs of models: $\tau = (C - D) / \binom{N}{2}$. We can then rewrite decision accuracy defined only by the number of concordant pairs C : decision accuracy = $C / \binom{N}{2}$.

Since we do not allow ties, C and D make up the total number of pairs $\binom{N}{2} = C + D$, we can rewrite¹⁰ decision accuracy as follows:

$$\begin{aligned} \tau &= \frac{C - \left(\binom{N}{2} - C\right)}{\binom{N}{2}} = \frac{2C - \binom{N}{2}}{\binom{N}{2}} = 2 \cdot \frac{C}{\binom{N}{2}} - 1 \\ &= 2 \cdot (\text{decision accuracy}) - 1 \end{aligned} \quad (11)$$

Therefore, the decision accuracy measure in Magnusson et al. [38] is equivalent to Kendall’s τ ¹² modulo a scale and shift.

Spearman’s Rank Correlation. Kendall’s τ is not sensitive to outliers, and instead we can incorpo-¹³ rate the strength of the difference in rank with Spearman’s ρ : $\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$. This statistic will be more sensitive to large differences in model ranking.

We use decision accuracy in this work for consistency, and to provide a more interpretable metric of¹⁴ rank agreement (for instance, a decision accuracy of 80% indicates that 80% of the pairs of mixes agree between the small scale and large scale). To show that both additional measures of agreement produce similar conclusions, we include correlation with these additional measures of agreement in Table 3.

A.3 Measures of Modeling Noise¹⁵

Seed Noise. To measure the noise introduced from changing the random seed initialization between¹⁶ training runs, we can compute the standard deviation of the final checkpoint from multiple training runs with different random seeds. To estimate seed noise, we train M models using the same configuration, and average the scores over the final n checkpoints of T total training checkpoints to smooth the checkpoint-to-checkpoint noise, then compute the standard deviation:

$$\text{Seed Noise}(M) = \sigma(M), \quad M_i = \frac{1}{n} \sum_{j=T-n+1}^T U(t_j) \quad (5)$$

Data Order Noise. This is noise introduced from changing the order of sampled documents from the training data. We estimate the data order noise using the same method as seed noise. 1

Total Variation. To measure the checkpoint-to-checkpoint noise throughout an entire training run, we measure the total variation of the intermediate training checkpoints on the downstream benchmark. We measure total variation as the average change in metric score across T training checkpoints minus an improvement term: 2

$$\text{Total Variation} = \frac{1}{T} \sum_{t=1}^T |U(t) - U(t-1)| - \frac{1}{T}(U(T) - U(0)) \quad 3 \quad (6)$$

Checkpoint-to-checkpoint Noise. Calculating the above sources of noise are either too expensive to estimate at large scales (e.g., training LLMs by varying the random seed) or difficult to run (e.g., evaluating every checkpoint on an LLM training curve). Instead, we propose an estimate measuring only the noise of the final n training checkpoints of training: 4

$$\text{Checkpoint-to-checkpoint Noise} = \sigma \left(\{U(t_j)\}_{j=T-k+1}^T \right) \quad 5 \quad (7)$$

A.3.1 Correlation between Sources of noise 6

To measure the relationship between each source of noise, we train 10 1B-5xC models varying the random seed initializations and 10 models varying the data order. In Figure 7, we measure the correlation between the seed noise, data order noise and total variation against the step-to-step noise. Each source of noise is highly correlated with the step-to-step noise ($R \geq 0.9$ for all measures). While it would be ideal to calculate and reduce all sources of noise, seed noise and data order noise are too expensive to measure (e.g., for large model runs as in Madaan et al. [36]), so only calculating step-to-step noise is a reasonable estimate for the modeling noise. Thus, we use step-to-step noise in as our estimate of the modeling noise. 7

A.3.2 Selecting the Number of Checkpoints in Noise 8

The noise calculation introduced in Section 3.1 requires selecting some n intermediate checkpoints to estimate the checkpoint-to-checkpoint noise. In this section, we provide guidance on selecting n , and discuss its impact on our findings. Increasing the number of intermediate checkpoints n will lead to a less biased estimate of noise. Thus, we can calculate the minimum number of n intermediate checkpoint samples such that the sample noise s_n is a reasonable estimate of the population noise σ . 9

We first assume the checkpoint to checkpoint scores are independent and normally distributed (which we observe when computing decision accuracy on intermediate checkpoints in Figure 7). Under this assumption, the ratio between the sample variance and the population variance follows a scaled chi squared distribution: $\frac{(n-1)s_n^2}{\sigma^2} \sim \chi_{n-1}^2$ 10

Therefore we would like to calculate the probability that the sample standard deviation s_n is within one standard deviation of the population standard deviation σ : $|s_n - \sigma| < \sigma$ 11

We can rewrite this inequality: 12

$$\left| \frac{s_n}{\sigma} - 1 \right| < 1 \Rightarrow 0 < \frac{s_n}{\sigma} < 2 \quad 13$$

And then, can substitute the chi-squared distribution to compute the likelihood w.r.t. n : 14

$$\frac{s_n}{\sigma} \sim \sqrt{\frac{\chi_{n-1}^2}{n-1}} \Rightarrow P \left(\sqrt{\frac{\chi_{n-1}^2}{n-1}} < 2 \right) \Rightarrow P(\chi_{n-1}^2 < 4(n-1)) \quad 15$$

We can then solve the inequality for the smallest value of n for a particular threshold α : 16

$$P(\chi_{n-1}^2 < 4(n-1)) > \alpha \quad 17$$

Solving this inequality numerically with $\alpha = 0.95$ for increasing values of n , we find that $n = 9$ provides the smallest sample size such that the probability that the sample standard deviation (the 18

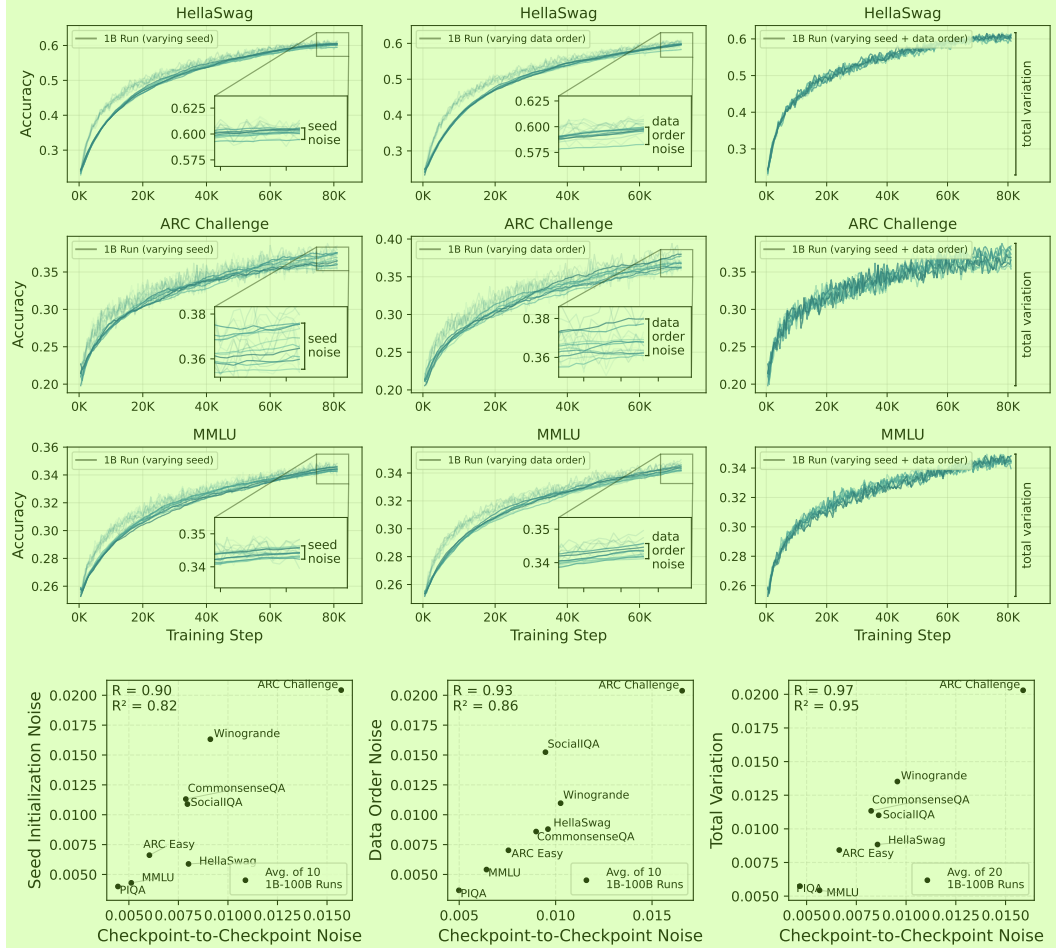


Figure 7: **Top:** 10 different training runs (1B-5 $\times$ C scale) varying random seed initialization and data order, plotting ARC-C accuracy smoothed across a window of 20 checkpoints. **Bottom:** Total variation or the relative standard deviation (STD normalized by average performance; §3) of scores from different seeds, data after averaging the last 20 training checkpoints vs. the Rel. Std. over the last 20 training checkpoints. Benchmarks with a high checkpoint-to-checkpoint noise also exhibit high noise due to random seed initialization, data order and noise along the full training curve. Noise for all tasks reported in Figure 19.

observed noise) is within one standard deviation of the population standard deviation (the true noise) with 95% confidence. In addition, we can specify a stricter bound by defining the sample standard deviation to be within $k \cdot \sigma$ of the population standard deviation: $|s_n - \sigma| < k \cdot \sigma$

We then verify this empirically using our estimate for noise at the 7B scale (from §5.2). If we assume the 30 intermediate checkpoints provide a reasonable estimate of the population standard deviation, we then compute the sample standard deviation s_n for $n < 30$. We re-compute s_n 1000 times for different subsets to calculate the likelihood that the sampled standard deviation is within $k \cdot \sigma$ of the population standard deviation σ . In the below table, we report this likelihood with tolerances $k \in \{0.2, 1.0\}$ for subsets $n \in \{5, 10, 20\}$ and bold all results with a likelihood above 0.95.

In practice, we find that for a large bound (± 1 std. dev.) can be satisfied for almost all benchmarks with $n = 5$ intermediate checkpoints, but for smaller bounds, (20% of ± 1 std. dev.), using $n = 20$ gives an adequate estimate for 34 of 39 benchmarks we considered in our work.

For our experiment on the 1B-5 $\times$ C checkpoints, we estimate noise using the average noise of the last 5 checkpoints for all 25 models, so our estimate of noise considers $5 \cdot 25 = 125$ scores.

Table 2: Ablating the n term in noise: Likelihood that the sample standard deviation for n intermediate checkpoints is a reasonable estimate for the population standard deviation on OLMo 2 7B, calculated using 30 intermediate checkpoints (Values for $\alpha > 0.95$ in bold). We find that for a low tolerance (within 0.2σ), 20 intermediate checkpoints provides an adequate estimate of noise.

k threshold in $k \cdot \sigma \rightarrow$ # Ckpts in Noise (n) $\rightarrow$	$k = 0.2$			$k = 1.0$		
	5	10	20	5	10	20
AGI Eval	0.42	0.61	0.95	1.00	1.00	1.00
ARC Challenge	0.44	0.70	0.98	1.00	1.00	1.00
ARC Easy	0.38	0.65	0.97	1.00	1.00	1.00
AutoBench	0.47	0.71	0.97	1.00	1.00	1.00
BBH	0.42	0.60	0.95	1.00	1.00	1.00
BoolQ	0.16	0.45	0.88	1.00	1.00	1.00
HumanEval	0.52	0.79	0.99	1.00	1.00	1.00
HumanEval+	0.47	0.76	0.99	1.00	1.00	1.00
CommonsenseQA	0.39	0.64	0.96	1.00	1.00	1.00
DROP	0.48	0.76	0.99	1.00	1.00	1.00
GSM8K	0.49	0.77	0.99	1.00	1.00	1.00
GSM+	0.50	0.79	0.99	1.00	1.00	1.00
GSM Symbolic	0.37	0.64	0.96	1.00	1.00	1.00
GSM Symbolic P1	0.47	0.69	0.98	1.00	1.00	1.00
GSM Symbolic P2	0.32	0.57	0.94	1.00	1.00	1.00
HellaSwag	0.39	0.65	0.97	1.00	1.00	1.00
Jeopardy	0.42	0.69	0.98	1.00	1.00	1.00
MBPP	0.43	0.63	0.96	1.00	1.00	1.00
MBPP+	0.41	0.63	0.96	1.00	1.00	1.00
MedMCQA	0.50	0.79	0.99	1.00	1.00	1.00
Minerva MATH	0.38	0.53	0.93	1.00	1.00	1.00
Minerva MATH 500	0.28	0.53	0.92	1.00	1.00	1.00
MMLU	0.00	0.00	0.54	0.83	1.00	1.00
MMLU Pro	0.51	0.78	0.99	1.00	1.00	1.00
All Tasks	0.00	0.00	0.08	0.83	1.00	1.00
Code Tasks	0.49	0.78	0.99	1.00	1.00	1.00
Knowledge Tasks	0.00	0.00	0.15	0.83	1.00	1.00
Math Tasks	0.55	0.83	0.99	1.00	1.00	1.00
OLMES Core 9	0.31	0.49	0.92	1.00	1.00	1.00
OLMES Gen	0.48	0.74	0.98	1.00	1.00	1.00
OpenBookQA	0.42	0.73	0.98	1.00	1.00	1.00
PIQA	0.43	0.69	0.98	1.00	1.00	1.00
SocialIQA	0.30	0.44	0.88	0.99	1.00	1.00
SQuAD	0.48	0.72	0.99	1.00	1.00	1.00
TriviaQA	0.48	0.76	0.99	1.00	1.00	1.00
WinoGrande	0.42	0.67	0.97	1.00	1.00	1.00

A.4 Measures of Signal

Measurements. When designing an measure of signal, we want to incorporate the uniformity of benchmark scores and the overall range of scores. Given the final checkpoints of training runs under similar compute spend C_{final} , we evaluate multiple approaches to measuring signal:

- **Variance** measures average squared distance from the mean: $\text{Var}(C_{\text{final}}) = \frac{1}{n} \sum_{i=1}^n \|c_i - \bar{c}\|^2$
- **Mean distance** measures average pairwise distance between points: $\text{Mean Dist}(C_{\text{final}}) = \frac{2}{n(n-1)} \sum_{i < j} \|c_i - c_j\|$
- **Relative standard deviation**, or the coefficient of variation, measures the standard deviation divided by the mean: $\text{Rel. Std.}(C_{\text{final}}) = \frac{\sqrt{\text{Var}(C_{\text{final}})}}{\text{Mean}(C_{\text{final}})}$
- **Star Discrepancy** measures the largest difference between any point and the uniform distribution: $\text{Discrepancy}(C_{\text{final}}) = \sup_{t \in [0,1]} \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{c_i \leq t\} - t \right|$.
- **Dispersion** measures the largest difference between any two points, or the largest unfilled space in the range of performance: $\text{Dispersion}(C_{\text{final}}) = \max_{i \neq j} \|c_i - c_j\|$.

Note, we include metrics that are sensitive and non sensitive to outliers, and find our results hold when measuring both types of spread (Table 3). We also include variants of these terms, such using a min-max normalization or scaling by the mean.

Choosing the a signal measurement. In Table 3, we calculate the correlation between signal-to-noise ratio and decision accuracy when using each of the signal variants. We see that many

Table 3: Correlation of signal-to-noise ratio to decision accuracy, using different measures of signal. We use the measure which is most predictive of decision accuracy as our measure of signal. We include alternative methods for calculating decision accuracy (Pearson correlation and Spearman’s rank correlation coefficient), as detailed in Appendix A.2. Fits are illustrated in Figure 10.

Measure of Signal		SNR vs. Decision Acc R^2	SNR vs. Pearson R^2	SNR vs. Spearman R^2
Rel. Dispersion	$\max_{i,j} c_i - c_j /\bar{c}$	0.5687	0.4052	0.4902
Rel. Std. Dev.	σ/μ	0.5657	0.3850	0.4771
Rel. Mean Pairwise Distance	$\frac{1}{n^2} \sum_{i,j} c_i - c_j /\bar{c}$	0.5458	0.3624	0.4561
Interquartile Range	$Q_3 - Q_1$	0.4836	0.2866	0.3980
Distance Standard Deviation	$\frac{1}{n} \sum_i (c_i - \bar{c})$	0.4745	0.3667	0.3950
RMS Deviation	$\sqrt{\frac{1}{n} \sum_i (c_i - \bar{c})^2}$	0.4633	0.3435	0.3812
Mean Pairwise Distance	$\frac{1}{n^2} \sum_{i,j} c_i - c_j $	0.4589	0.3325	0.3758
Range	$\max(c) - \min(c)$	0.4574	0.3604	0.3865
Dispersion	$\max_{i,j} c_i - c_j $	0.4574	0.3604	0.3865
Quartile Deviation	$(Q_3 - Q_1)/2$	0.4528	0.2896	0.3655
Average Absolute Deviation	$\frac{1}{n} \sum_i c_i - \bar{c} $	0.4507	0.3186	0.3672
Median Absolute Deviation	$\text{median}(c_i - \text{median}(c))$	0.4168	0.2663	0.3346
Rel. Mean Squared Pairwise Distance	$\frac{1}{n^2} \sum_{i,j} (c_i - c_j)^2/\bar{c}^2$	0.2908	0.1627	0.2324
Mean Squared Pairwise Distance	$\frac{1}{n^2} \sum_{i,j} (c_i - c_j)^2$	0.2480	0.1457	0.1953
Gini Coefficient	$\frac{1}{2n^2\mu} \sum_{i,j} c_i - c_j $	0.0944	0.0978	0.0829
Star Discrepancy (Shift+Scale)	$\sup_{[0,c]} F_n(t) - F(t) $ with shifting	0.0391	0.0768	0.0454
Star Rel. Discrepancy	$\sup_{[0,c]} F_n(t) - F(t) /F(t)$	0.0379	0.0587	0.0420
Dispersion (Shift+Scale)	$\max_{i,j} c_i - c_j $ with shifting	0.0374	0.0679	0.0382
Halfspace Depth	$\min(F_n(x), 1 - F_n(x))$	0.0358	0.0395	0.0373
Discrepancy	$\max_c F_n(c) - F(c) $	0.0340	0.0754	0.0401
Projection Depth	$\left(1 + \frac{ x - \text{med}(c) }{\text{MAD}(c)}\right)^{-1}$	0.0331	0.0392	0.0353
Star Discrepancy	$\sup_{[0,c]} F_n(t) - F(t) $	0.0319	0.0665	0.0356

straight forward measures have similarly high correlations. We use relative dispersion, the highest correlated among them, as our measure of signal.

A.5 Dataset Details

A.5.1 Models

We evaluate 465 models which represent stages of the decision-making process during pre-training. Unlike existing collections of model evaluations [18, 33], our set is targeted at development models:

Scaling Law Models. 25 ladder models from Bhagia et al. [3]. $\{190\text{M}, 370\text{M}, 760\text{M}, 1.3\text{B}, 3.2\text{B}\} \times \{0.5\times\text{C}, 1\times\text{C}, 2\times\text{C}, 5\times\text{C}, 10\times\text{C}\}$ trained on OLMoE mix, and 7B-4T / 13B-5T as prediction targets.

Decision Accuracy Models. 225 models from Magnusson et al. [38] trained on 25 data recipes for $\{4\text{M}, 20\text{M}, 60\text{M}, 90\text{M}, 150\text{M}, 300\text{M}, 530\text{M}, 750\text{M}, 1.3\text{B}\}$ trained to 5x Chinchilla optimal.

Random Seed & Data Order Models. 20 models 1B-5xC models trained on the OLMoE mix, 10 models trained with different random seed initializations and 10 models trained with different data order seeds.

Final n Checkpoints. 120 models representing the 30 final checkpoints before the end of training for OLMo 2 1B, 7B, 13B and 32B [42], with checkpoints spaced by 1000 training checkpoints.

External Models. 73 open-weight base models from the DCLM, DeepSeek, Gemma, Llama, Orca, Phi, Pythia, Qwen, SmoLLM, StableLM and Yi model families. We estimate the training FLOPs using the reported token count.

We perform all evaluation using up to 2 H100s for a particular model, and use 94K H100 hours total for all evaluation. For training our randomly initialized seed and data order models, we use 23K GPU hours, using a cluster of 2x8 H100s for each training run.

A.5.2 Benchmarks

We intentionally select benchmarks that are widely adopted in pre-training evaluation. We use the OLMES [22] standard when applicable, and for other benchmarks, we reproduce the evaluation setup

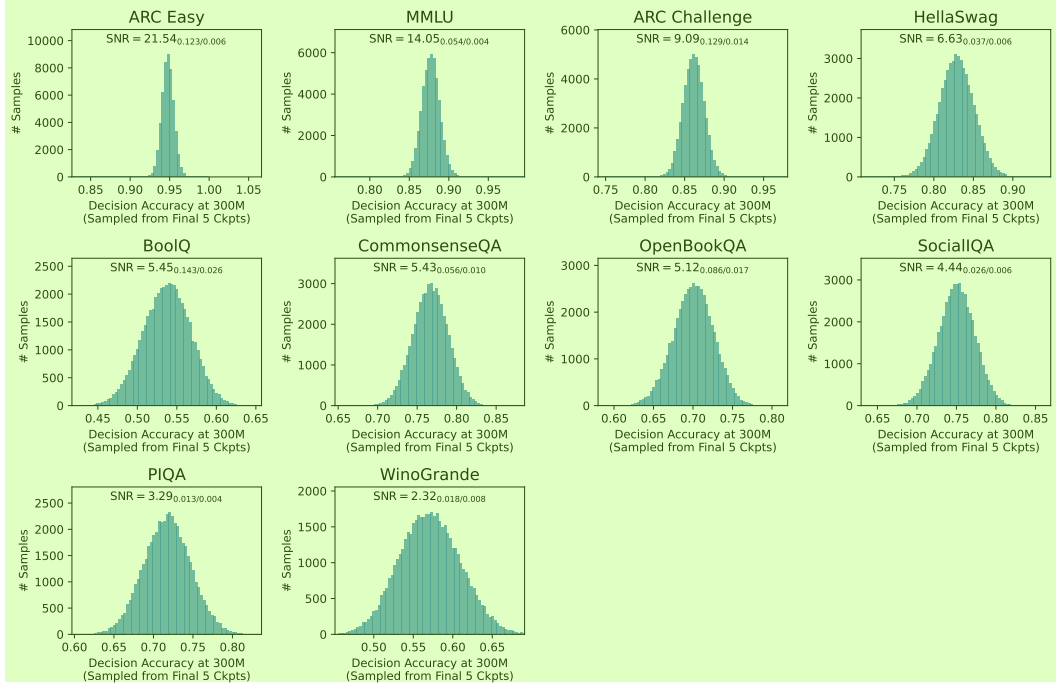


Figure 8: As the benchmark’s signal-to-noise ratio increases (across histograms), decision accuracy (from 300M to 1B scale) not only increases but becomes more consistent. We test this by resampling decision accuracy for combinations among last 5 checkpoints of the small and large models, respectively, since noise in the results of either size can change rankings. Note how CSQA and MMLU have similar signal (Rel. Dispersion = 0.056 vs 0.054) but different noise (Rel. Std. = 0.01 vs. 0.004).

from OLMo 2 [42]. Notably, all tasks use few-shot examples and we evaluate MCQA benchmarks in both the rank choice (RC) and multiple choice (MC) setting, since our small ($\leq 1B$ parameter) models show random-chance performance on MCQA benchmarks.

Knowledge QA. MMLU [23], ARC [11], BoolQ [10], CSQA [59], OBQA [39], PiQA [4], SocialQA [53], HellaSwag [69], WinoGrande [52], DROP [16], CoQA [48], Jeopardy [61], NaturalQs [28], SQuAD [47], TriviaQA [26], MedMCQA [43], MMLU Pro [64], AGI Eval [70], GPQA [49]

Math. GSM [12], GSM Plus [46], GSM Symbolic [41], Minerva [30]

Code. HumanEval [8], HumanEval+ [34], MBPP [1], MBPP+ [34]

Using strong LLMs have become a tool for augmenting existing benchmarks with more difficult questions or answer choices [64] and re-evaluating benchmark quality [62], and may provide a cheap method for improving signal. To test this, we add an additional synthetic benchmark:

Autobench. To test whether fully generated benchmarks can act as an adequate development benchmark, we generate a dataset of 30K MCQA questions using Autobench [32]. Autobench iteratively mines for Wikipedia articles and uses a strong LM to generate and prune questions based on saliency, novelty and difficulty constraints.

B Full Results

B.1 Noise measures the reliability of decision accuracy.

As discussed in §3.1, the checkpoint-to-checkpoint noise can change the ranking of models, which may effect the decision accuracy we observe by only evaluating the final DataDecide model. To measure the impact of checkpoint-to-checkpoint noise on decision accuracy, we can estimate the distribution of possible decision accuracies given the step to step noise. To do this, we sample one of the final 5 checkpoints for both the small and large model, and repeatedly sample to estimate the

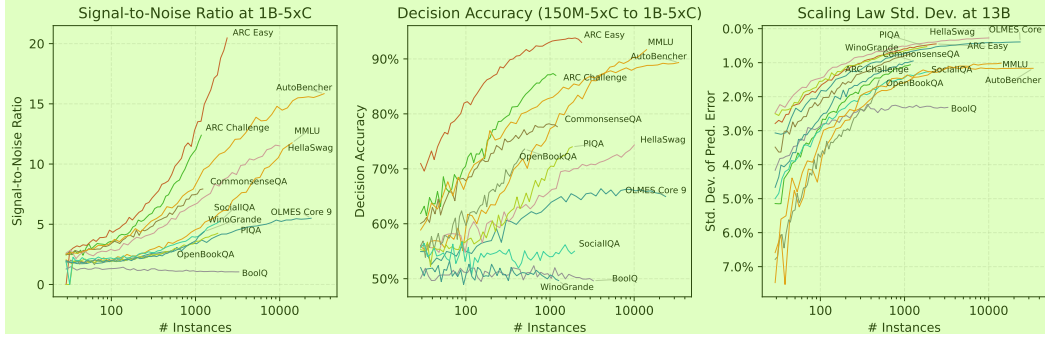


Figure 9: Signal-to-noise ratio, decision accuracy, and scaling law prediction error for randomly sampled subsets of instances for 6 development benchmarks. A large sample size alone does not improve signal-to-noise ratio. For example, a 1000 question subset of ARC Easy has a higher decision accuracy than MMLU despite having 90% fewer instances.

distribution. A wider distribution would indicate that one should be less confident in the decision accuracy.

We show the distribution of decision accuracies for 10K random samples in Figure 8. For tasks with a higher signal-to-noise ratio, the sampled decision accuracy distribution has a higher mean and lower variance. Additionally, we find that tasks with similar signal, but different noise (e.g., CSQA and MMLU, where CSQA has higher noise), the tasks with lower noise also have a lower variance of sampled decision accuracy distribution.

B.2 Increasing benchmark size has diminishing returns

Setup. One intuitive way to reduce modeling noise is to increase the size of the benchmark, while this is expensive in practice, recent work has given LLMs access to privileged information to generate distractor options or full benchmarks [32, 64]. To test the impact of sample size on modeling noise, we use the existing set of benchmarks, select a random sample of instances and recalculate SNR, decision accuracy and scaling law error. To test the limits of synthetic benchmarks, we use our version of AutoBench, which has 33K instances, or 2x more test instances than the next largest benchmark in our dataset (MMLU).

Results. Figure 9 shows how each metric improves as the number of instances increases. Initially, all benchmarks benefit from more samples (up until ~1K samples) as expected. However, we find diminishing returns for some benchmarks after only 1K instances, in particular the signal-to-noise ratio for AutoBench shows an inflection point at around 2K instances. This is due to the AutoBench having high noise, as shown by the scaling law standard deviation (right figure) – despite having the largest sample size, AutoBench has the highest checkpoint-to-checkpoint noise. In fact, the 300 instance subset of ARC-Easy has lower noise than the full 30K instance AutoBench. As using LLMs as part of benchmark construction has become a more popular method of constructing benchmarks, a high quality, small benchmark can actually show a less noisy signal.

B.3 Signal-to-Noise Ratio at Large (>32B) Scales

Setup. For models larger than the DataDecide scale (1B-100B), we can rely on the signal-to-noise ratio directly to indicate development benchmarks which may not be useful. We estimate the signal-to-noise ratio at the compute scales used to train the OLMo 2 models: 1.5B-4T, 7B-4T, 13B-5T and 32B-6T. For noise, we use the final 30 intermediate checkpoints, one checkpoint for every 1000 training steps until the end of training. For signal, we do not have access to different data recipes trained on the same model, so instead we use a population of open-weight base models trained to similar compute budget as the OLMo 2 models. We use models trained using $\pm 10\%$ of the estimated FLOPs, which results in a population of at least 8 models for each size.

Results. Table 4 reports the SNR for each compute budget, sorted by SNR at the 1.5B-4T model scale. SNR can indicate when benchmarks saturated, for example ARC Easy and SocialQA have high SNR at 1.5B-4T, but low SNR at 32B-6T: 7.89 to 5.10 and 8.73 to 1.95 respectively. For these

Table 4: Signal-to-noise ratio for language model development benchmarks for the compute budgets of the OLMo 2 family [42]. For benchmarks measuring a similar ability, we recommend using benchmarks with a higher signal-to-noise ratio for a particular model scale. Performance on all models is shown in Figure 12.

Model Size → Compute → Benchmark ↓	1.5B-4T $2 \cdot 10^{22}$ FLOPs SNR _{Signal/Noise}	7B-4T $1.6 \cdot 10^{23}$ FLOPs SNR _{Signal/Noise}	13B-5T $3.9 \cdot 10^{23}$ FLOPs SNR _{Signal/Noise}	32B-6T $1.2 \cdot 10^{24}$ FLOPs SNR _{Signal/Noise}
Knowledge QA Tasks				
HellaSwag	39.77 _{0.180/0.005}	23.94 _{0.061/0.003}	17.81 _{0.054/0.003}	8.20 _{0.028/0.003}
TriviaQA	28.15 _{0.411/0.015}	47.03 _{0.135/0.003}	60.37 _{0.141/0.002}	27.19 _{0.064/0.002}
Jeopardy	23.66 _{0.374/0.016}	14.38 _{0.082/0.006}	18.49 _{0.084/0.005}	8.00 _{0.032/0.004}
OLMES Gen	19.34 _{0.247/0.013}	32.58 _{0.129/0.004}	4.19 _{0.092/0.022}	1.06 _{0.048/0.046}
OLMES Core 9	19.11 _{0.118/0.006}	9.61 _{0.039/0.004}	7.13 _{0.030/0.004}	8.16 _{0.027/0.003}
AutoBench	17.62 _{0.264/0.015}	11.42 _{0.102/0.009}	8.23 _{0.105/0.013}	3.73 _{0.050/0.014}
MMLU Pro	16.28 _{0.246/0.015}	17.44 _{0.168/0.010}	9.34 _{0.098/0.010}	15.04 _{0.136/0.009}
MMLU	14.52 _{0.139/0.010}	3.39 _{0.078/0.023}	7.51 _{0.044/0.006}	5.19 _{0.061/0.012}
PIQA	14.23 _{0.058/0.004}	5.31 _{0.023/0.004}	5.52 _{0.023/0.004}	4.97 _{0.015/0.003}
WinoGrande	14.12 _{0.118/0.008}	7.35 _{0.062/0.008}	7.68 _{0.070/0.009}	6.60 _{0.046/0.007}
CommonsenseQA	12.17 _{0.120/0.010}	5.66 _{0.033/0.006}	2.69 _{0.022/0.008}	7.05 _{0.039/0.006}
DROP	10.79 _{0.337/0.031}	20.79 _{0.262/0.013}	12.19 _{0.226/0.019}	9.01 _{0.143/0.016}
ARC Challenge	9.41 _{0.193/0.021}	5.85 _{0.081/0.014}	2.32 _{0.033/0.014}	4.74 _{0.064/0.014}
SocialQA	8.73 _{0.119/0.014}	5.15 _{0.049/0.010}	1.69 _{0.020/0.012}	1.95 _{0.026/0.013}
MedMCQA	8.59 _{0.106/0.012}	5.79 _{0.051/0.009}	7.70 _{0.060/0.008}	4.00 _{0.041/0.010}
ARC Easy	7.89 _{0.102/0.013}	5.77 _{0.035/0.006}	3.94 _{0.018/0.004}	5.10 _{0.018/0.004}
SQuAD	6.11 _{0.090/0.015}	9.76 _{0.061/0.006}	10.45 _{0.044/0.004}	3.92 _{0.027/0.007}
AGI Eval	5.31 _{0.105/0.020}	4.23 _{0.076/0.018}	2.74 _{0.050/0.018}	5.40 _{0.062/0.012}
BoolQ	4.87 _{0.116/0.024}	2.99 _{0.048/0.016}	1.18 _{0.016/0.013}	2.67 _{0.016/0.006}
OpenBookQA	4.82 _{0.145/0.030}	2.13 _{0.053/0.025}	2.42 _{0.048/0.020}	3.05 _{0.063/0.021}
Math Tasks				
GSM+	8.06 _{0.610/0.076}	13.07 _{0.500/0.038}	8.55 _{0.299/0.035}	8.42 _{0.199/0.024}
GSM Symbolic P1	7.18 _{0.831/0.116}	4.85 _{0.677/0.140}	6.54 _{0.450/0.069}	5.31 _{0.277/0.052}
GSM8K	3.83 _{0.587/0.153}	8.21 _{0.434/0.053}	6.98 _{0.255/0.037}	6.61 _{0.160/0.024}
GSM Symbolic P2	3.62 _{0.805/0.222}	2.98 _{0.769/0.258}	3.39 _{0.560/0.165}	4.67 _{0.468/0.100}
GSM Symbolic	3.05 _{0.662/0.217}	8.94 _{0.527/0.059}	6.61 _{0.283/0.043}	4.29 _{0.134/0.031}
Minerva MATH	2.28 _{0.568/0.250}	9.32 _{0.643/0.069}	7.48 _{0.567/0.076}	10.19 _{0.409/0.040}
Minerva MATH 500	0.91 _{0.491/0.539}	4.45 _{0.748/0.168}	4.44 _{0.647/0.146}	4.30 _{0.383/0.089}
Code Tasks				
HumanEval+	3.70 _{0.482/0.130}	7.18 _{0.432/0.060}	8.47 _{0.377/0.045}	3.34 _{0.131/0.039}
HumanEval	3.64 _{0.452/0.124}	6.25 _{0.395/0.063}	5.18 _{0.314/0.061}	3.19 _{0.117/0.037}
MBPP+	0.88 _{0.207/0.235}	3.60 _{0.302/0.084}	4.72 _{0.265/0.056}	2.94 _{0.137/0.047}
MBPP	0.88 _{0.221/0.251}	5.09 _{0.382/0.075}	4.52 _{0.255/0.057}	3.57 _{0.167/0.047}
Multi-task Averages				
Knowledge Tasks	17.70 _{0.146/0.008}	1.61 _{0.080/0.049}	9.82 _{0.048/0.005}	1.03 _{0.058/0.056}
OLMES + Gen	17.35 _{0.143/0.008}	2.65 _{0.074/0.028}	9.52 _{0.045/0.005}	0.93 _{0.052/0.056}
All Tasks	13.92 _{0.152/0.011}	3.68 _{0.128/0.035}	9.26 _{0.055/0.006}	2.94 _{0.075/0.026}
Math Tasks	5.78 _{0.656/0.113}	11.72 _{0.580/0.050}	5.06 _{0.384/0.076}	7.87 _{0.253/0.032}
Code Tasks	3.28 _{0.333/0.102}	8.20 _{0.371/0.045}	8.87 _{0.308/0.035}	5.55 _{0.126/0.023}

benchmarks, they have less powerful comparisons at larger sizes. SNR also indicates when particular benchmarks become useful. For example, Minerva MATH 500 has the lowest SNR of all tasks at 1.5B-4T (SNR = 0.91) but much higher SNR already at 7B-4T (SNR = 4.45).

Additionally, some individual tasks show better SNR than mutli-task averages. For the OLMES Core 9 average, HellaSwag has higher SNR at all model sizes. For OLMES Gen, TriviaQA has higher SNR at all model sizes. In cases where the SNR of the mutli-task average is low, like the OLMES Average, we recommend comparing models based on individual, high SNR tasks.

C Additional Results

We include for our core experiments across all benchmarks we study:

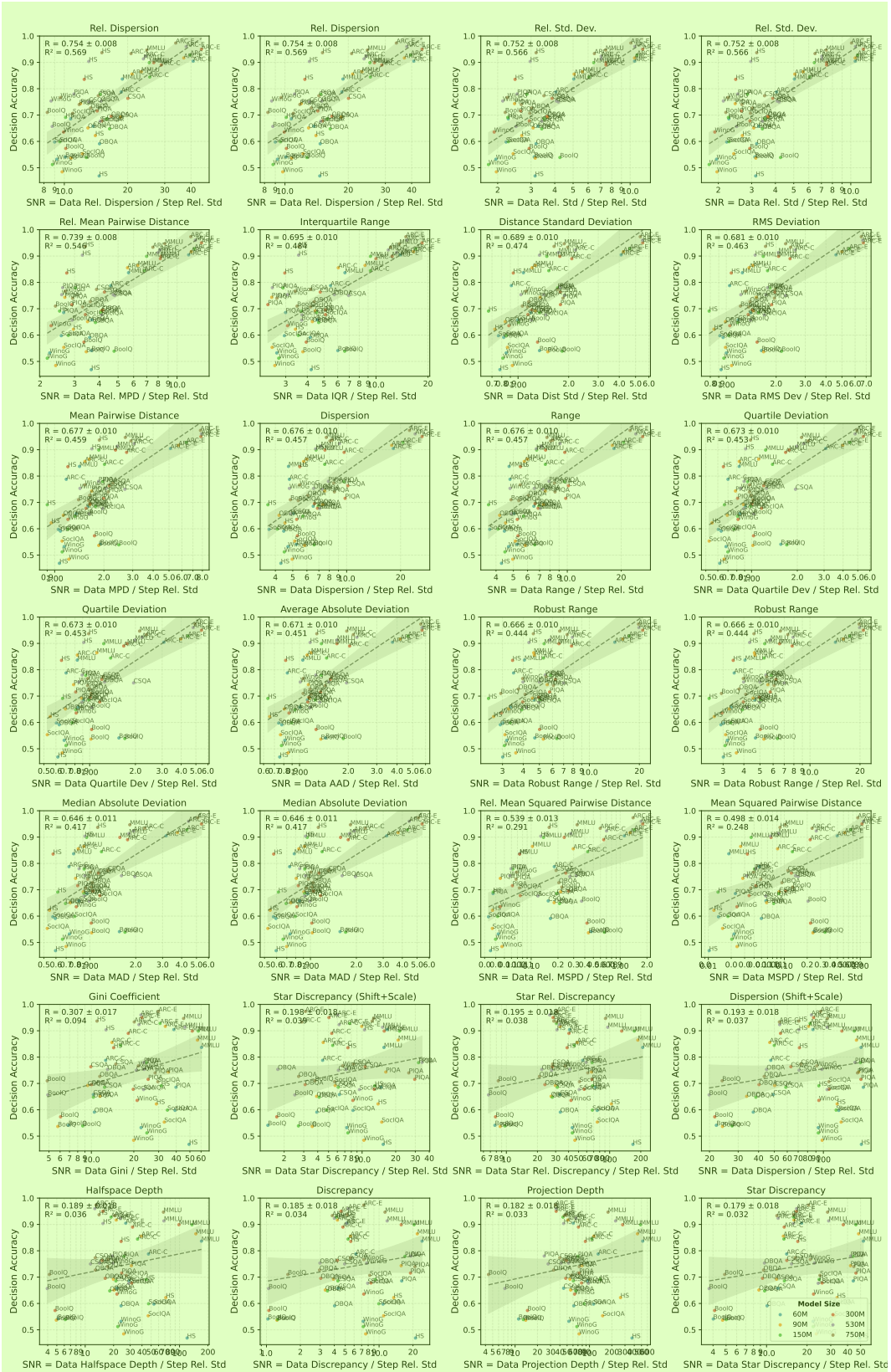


Figure 10: Correlation between decision accuracy and variants of signal-to-noise ratio, using 2 different measures of signal. To pick the measure of signal, we use the metric which is most predictive of decision accuracy.

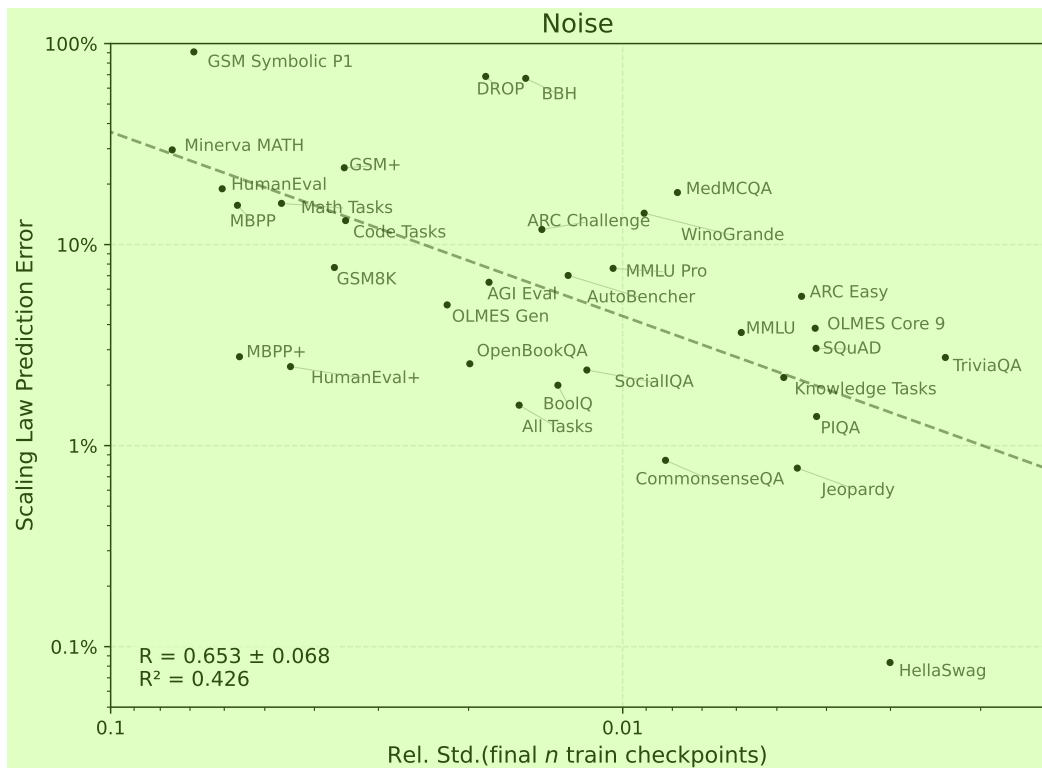


Figure 11: Scaled-up version of the Figure 3 in §4.2 with labels on each task. 2

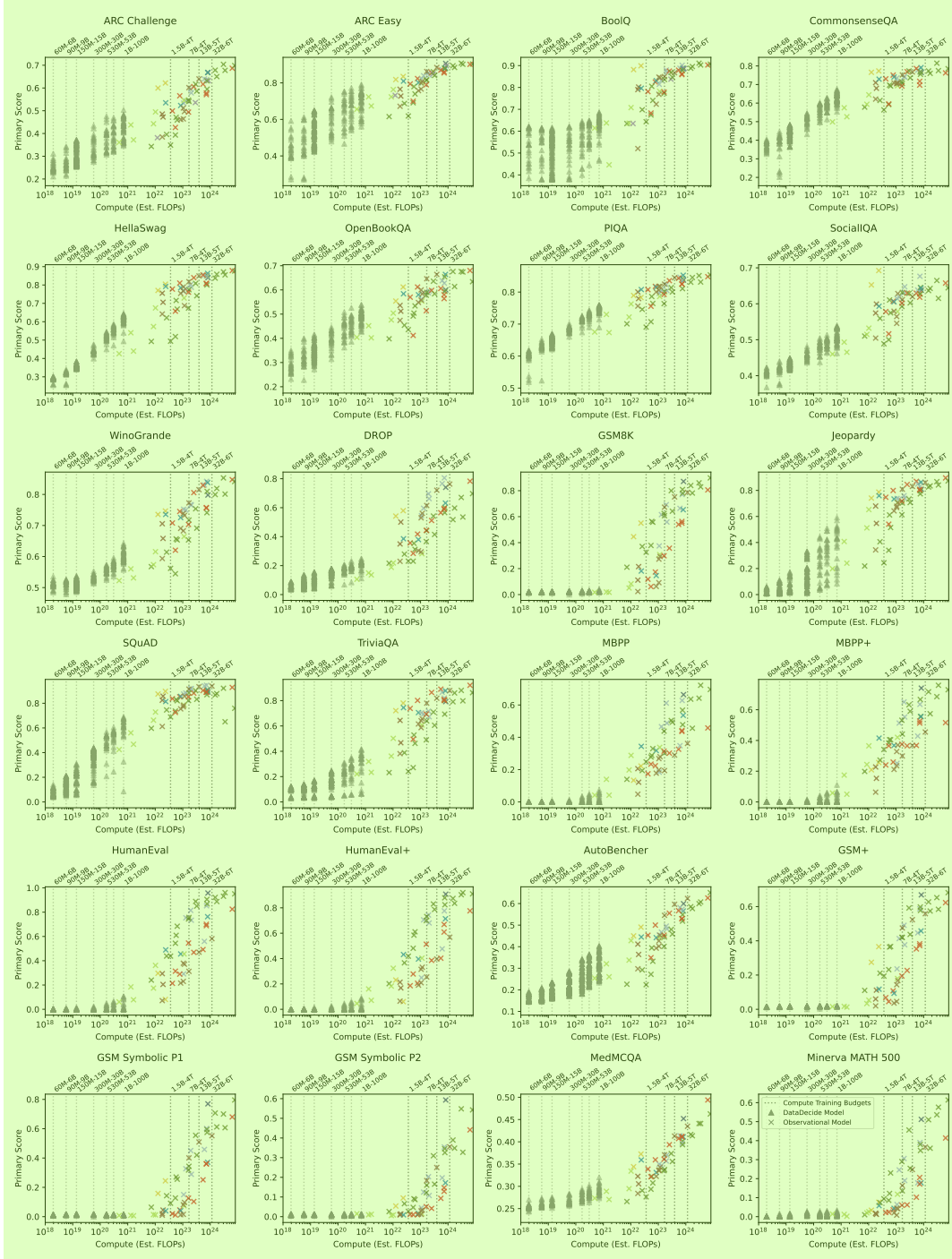


Figure 12: Performance of language models from 60M parameters to 32B parameters, which we use to measure spread at different training budgets in Table 4. For our core experiments, we use the DataDecide models to measure spread, and at large scales, we use external models trained at similar compute budgets.

1



Figure 13: Final 30 checkpoints, each spaced 1000 training steps, for OLMo 2 1B, 7B, 13B and 32B along with the Rel. Std. Dev., which is used to estimate noise.

2

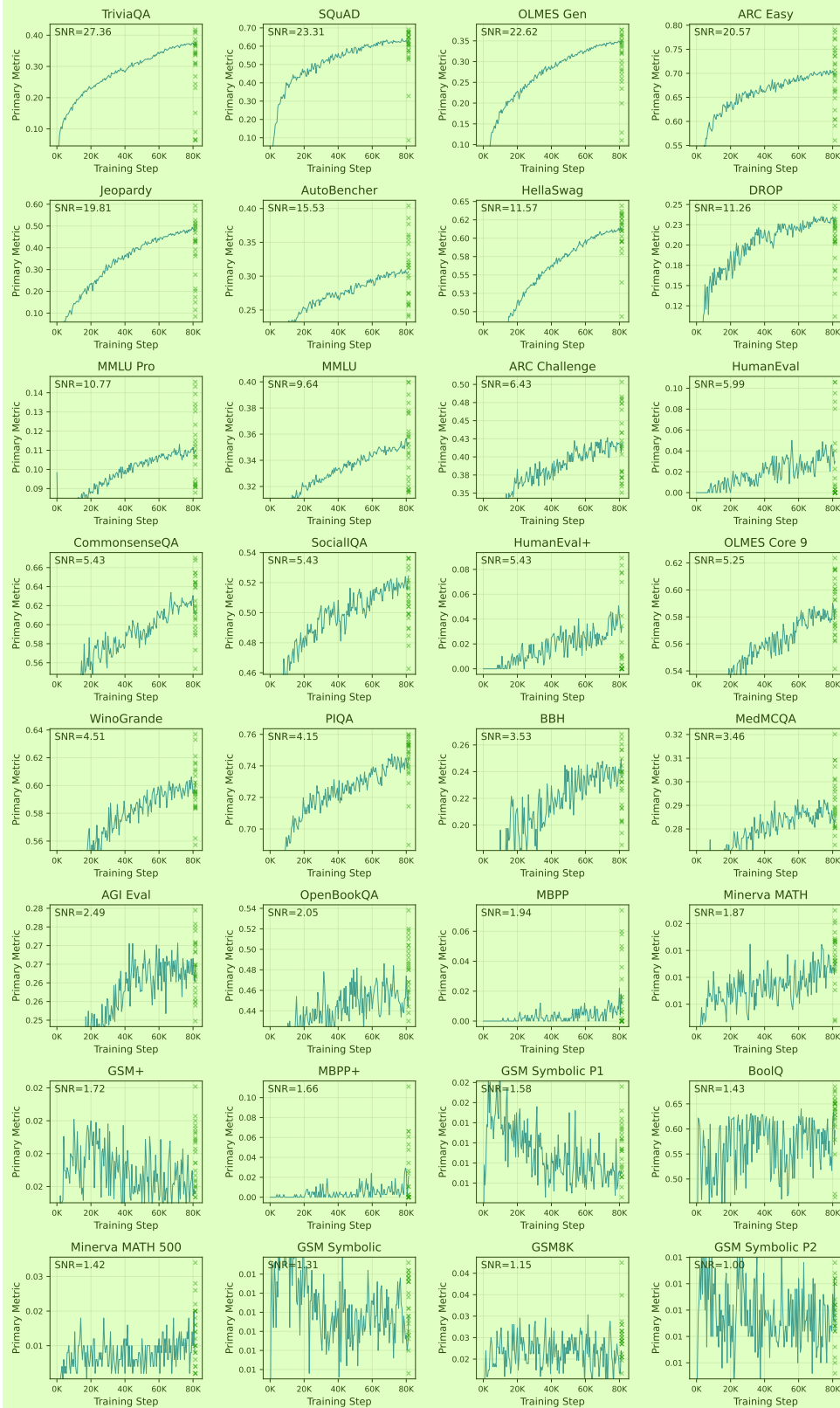


Figure 14: 1B-5xC training curves and final checkpoints for DataDecide models across tasks, sorted by the signal-to-noise ratio.

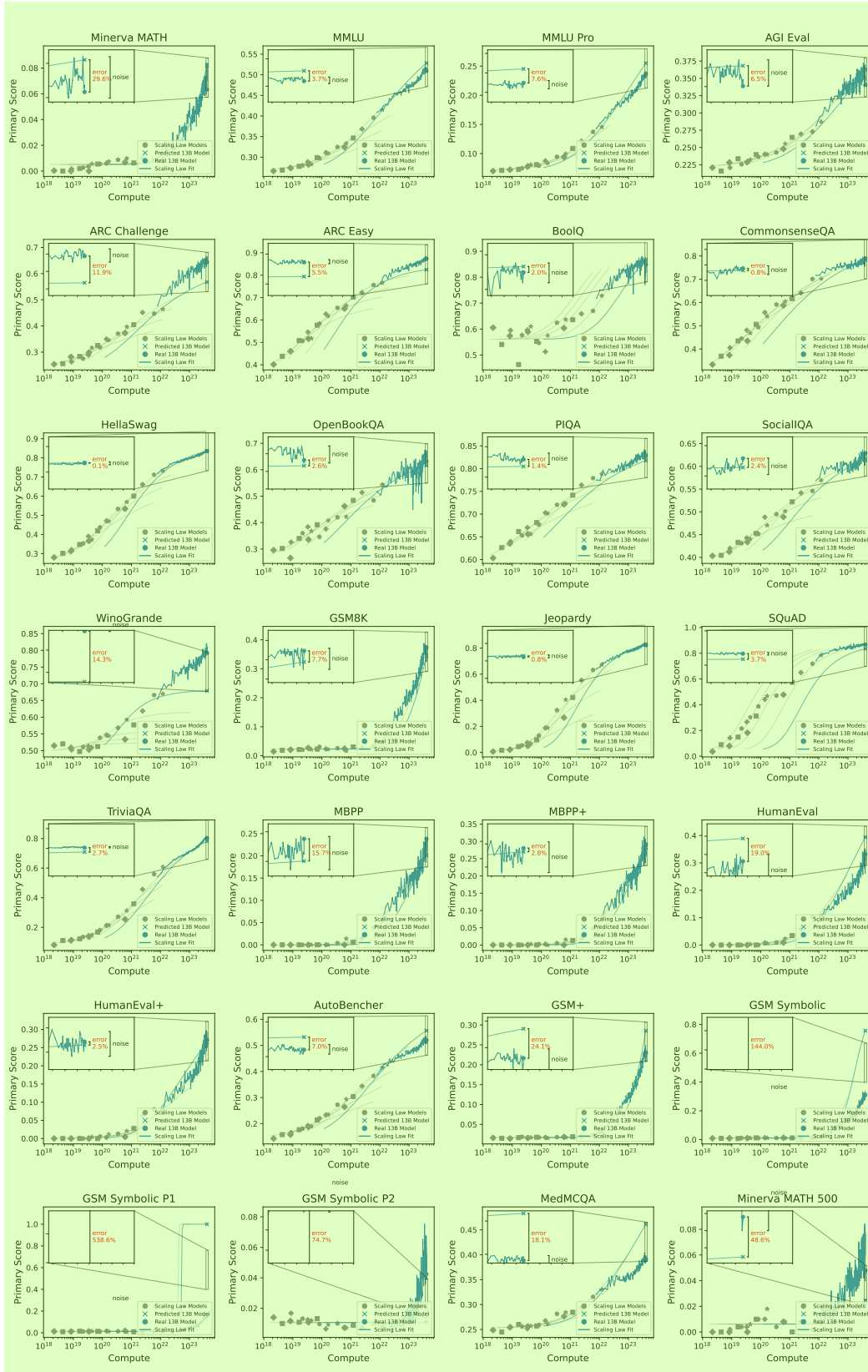


Figure 15: Scaling law fits for all tasks using the OLMo 2 13B-5T prediction target. 2

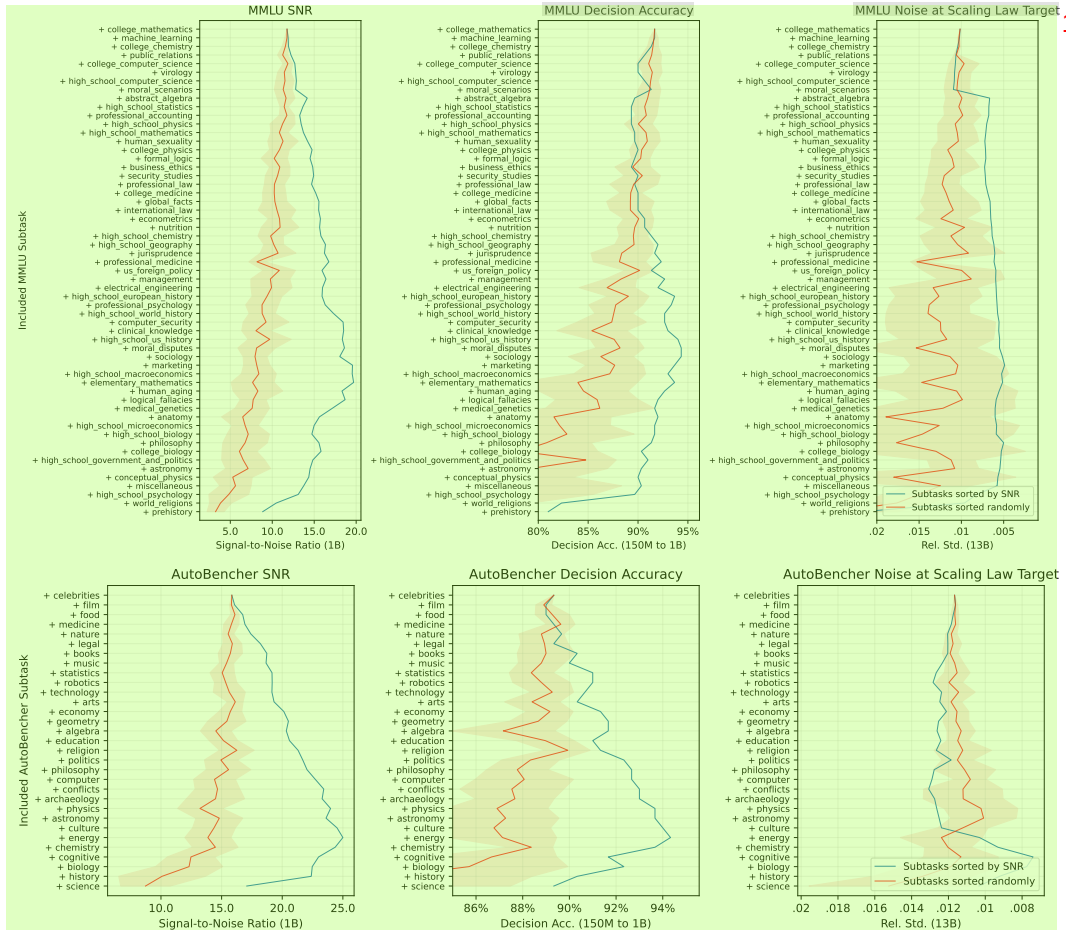


Figure 16: Larger version of Figure 4, showing the names of each subtask, sorted by SNR from bottom (highest SNR) to top (lowest SNR).

Table 5: Scaling law fit error for BPB and primary score for all tasks with averaging the final 5 checkpoints in the ladder train models.

Task (↓)	Predicting Bits-per-byte				Predicting Primary Score			
	Abs. Error, %		Rel. Error, %		Abs. Error, %		Rel. Error, %	
	Final Only	Avg. Train	Final Only	Avg. Train	Final Only	Avg. Train	Final Only	Avg. Train
Knowledge QA Tasks								
HellaSwag	0.76	0.80	1.16	1.22	0.31	0.16	0.37	0.20
CommonsenseQA	6.24	5.32	8.75	7.46	0.59	0.46	0.75	0.58
Jeopardy	5.08	5.14	18.51	18.73	0.57	0.54	0.69	0.66
SocialIQA	0.66	0.41	0.74	0.46	0.50	0.59	0.80	0.95
PIQA	1.23	1.39	1.40	1.59	0.89	1.01	1.08	1.22
MMLU	0.56	0.49	0.75	0.66	1.68	1.74	3.28	3.39
MMLU Pro	0.78	0.71	0.73	0.67	1.76	1.75	7.51	7.45
AGI Eval	2.79	2.66	3.33	3.18	1.89	1.98	5.43	5.70
OLMES Gen	4.66	2.32	3.92	1.95	4.19	2.16	6.22	3.20
BoolQ	1.49	1.76	8.54	10.11	4.13	2.48	4.91	2.96
OLMES Core 9	0.47	0.25	0.62	0.33	2.47	2.62	3.23	3.42
TriviaQA	1.56	2.05	2.27	2.98	2.33	2.62	2.89	3.25
SQuAD	4.96	4.96	32.35	32.37	2.80	2.79	3.23	3.21
OpenBookQA	3.18	3.92	2.80	3.46	4.02	3.38	6.22	5.22
AutoBench	2.92	2.78	4.70	4.49	3.86	3.69	7.47	7.14
ARC Easy	1.36	1.37	2.89	2.90	5.13	5.13	5.87	5.87
MedMCQA	5.07	5.38	5.35	5.67	7.72	7.98	19.72	20.41
ARC Challenge	2.08	2.07	3.15	3.14	8.44	8.43	13.02	13.01
WinoGrande	1.01	1.38	0.83	1.12	10.01	10.82	12.47	13.49
BBH	61.84	65.01	12.81	13.47	33.09	33.08	66.61	66.59
DROP	47.51	48.19	10.75	10.91	35.17	35.20	68.77	68.82
Knowledge 19-Task Avg.	1.18	0.87	1.32	0.98	1.43	1.20	2.22	1.85
Math Tasks								
Minerva MATH	0.73	0.66	1.50	1.36	1.08	0.98	15.28	13.93
Minerva MATH 500	0.34	0.14	0.71	0.29	17.35	1.78	306.18	31.36
GSM Symbolic P2	2.57	2.83	5.23	5.75	7.46	3.50	164.53	77.13
GSM8K	2.43	2.48	5.90	6.01	7.46	3.85	20.55	10.61
GSM+	2.02	1.95	4.54	4.40	29.14	28.54	130.01	127.36
GSM Symbolic	1.87	1.71	4.64	4.25	39.88	38.88	132.62	129.30
GSM Symbolic P1	2.31	2.35	5.04	5.11	27.15	83.62	178.46	549.63
Math 6-Task Avg.	2.05	2.01	4.52	4.42	11.33	2.30	65.52	13.28
Code Tasks								
HumanEval+	1.92	2.21	3.57	4.10	1.05	0.04	3.91	0.16
MBPP	0.30	0.32	0.46	0.48	2.57	1.79	11.63	8.10
MBPP+	6.49	6.62	12.56	12.81	9.08	8.79	33.14	32.11
HumanEval	1.59	2.01	3.85	4.87	7.71	8.85	24.00	27.55
Code 4-Task Avg.	3.23	3.33	6.07	6.25	3.15	2.75	11.61	10.15
All 30-Task Avg.	0.47	0.15	0.62	0.20	1.03	0.86	2.10	1.76

Table 6: Decision accuracy averaging the final 5 checkpoints for bits-per-byte and the primary metric (accuracy, exact match, pass@1).

Task (↓)	Final Ckpt	Bits-per-byte, %			Final Ckpt	Primary Metric, %		
		Avg. Pred	Avg. Target	Avg. Both		Avg. Pred	Avg. Target	Avg. Both
Knowledge QA Tasks								
ARC Challenge	94.56	94.88	94.38	94.67	82.91	82.27	82.91	82.00
HellaSwag	92.42	93.19	93.21	94.00	71.05	71.26	72.37	72.33
ARC Easy	92.23	92.15	91.96	92.00	93.96	93.99	94.05	94.00
MMLU	91.53	91.64	91.63	91.67	89.08	88.84	89.60	89.00
AutoBench	88.55	88.95	89.19	89.67	88.80	89.05	88.81	89.00
MMLU Pro	90.00	89.40	90.04	89.33	83.34	83.77	84.20	84.67
AGI Eval	86.38	86.75	86.54	87.00	57.38	58.60	56.45	57.67
MedMCQA	86.67	86.67	86.67	86.67	61.33	61.33	61.33	60.33
Jeopardy	84.42	84.46	84.88	85.00	83.01	82.60	83.74	83.33
TriviaQA	83.55	84.29	83.86	84.67	69.10	69.54	69.09	69.33
OpenBookQA	81.53	81.75	81.68	82.00	66.82	66.98	68.05	68.33
OLMES Core 9	79.05	80.10	79.32	80.33	74.67	73.92	74.24	73.67
SocialIQA	79.92	79.57	79.45	79.00	55.58	55.58	56.09	56.67
WinoGrande	73.20	74.29	72.83	74.00	50.52	50.27	49.81	49.00
PIQA	72.60	72.91	71.93	72.00	72.78	72.66	73.09	72.33
CommonsenseQA	65.86	66.25	65.42	65.67	68.74	69.05	70.61	71.00
BoolQ	63.72	64.19	63.51	64.00	50.38	48.90	50.66	49.33
SQuAD	60.93	60.59	62.02	61.67	58.69	58.35	59.72	59.33
OLMES Gen	61.16	55.44	55.11	58.86	62.06	54.87	53.42	50.12
DROP	56.67	56.48	57.46	57.33	57.77	59.06	57.80	59.33
BBH	57.48	57.25	57.66	57.33	59.15	59.88	60.85	61.33
Knowledge 19-Task Avg.	71.39	71.49	71.62	71.67	70.70	75.82	72.65	78.00
Math Tasks								
Minerva MATH 500	90.33	90.33	90.33	90.33	51.00	51.00	51.00	51.00
Minerva MATH	90.00	90.00	90.00	90.00	51.00	51.00	51.00	51.00
GSM Symbolic P1	81.33	81.33	81.33	81.33	41.67	41.67	41.67	41.67
GSM Symbolic P2	79.67	79.67	79.67	79.67	40.33	40.33	40.33	40.33
GSM+	79.00	79.00	79.00	79.00	59.67	59.67	59.67	59.67
GSM Symbolic	78.33	78.33	78.33	78.33	51.67	51.67	51.67	51.67
GSM8K	76.67	76.67	76.67	76.67	46.33	46.33	46.33	46.33
Math 6-Task Avg.	88.33	88.33	88.33	88.33	42.67	42.67	42.67	42.67
Code Tasks								
HumanEval+	96.33	96.33	96.33	96.33	71.33	71.33	71.33	71.33
HumanEval	95.67	95.67	95.67	95.67	80.00	80.00	80.00	80.00
MBPP	95.33	95.33	95.33	95.33	76.00	76.00	76.00	76.00
MBPP+	93.00	93.00	93.00	93.00	70.67	70.67	70.67	70.67
Code 4-Task Avg.	96.67	96.67	96.67	96.67	85.67	85.67	85.67	85.67
All 30-Task Avg.	68.57	70.63	69.78	71.33	62.15	68.88	67.29	77.33

Figure 17: Bits-per-byte vs. primary metric on the full suite of tasks shown in Figure 6. ¹

Experiment Setting → Metric →	SNR (↑)		Rel. Error (↓), %		Decision Acc (↑), %	
	Primary	BPB	Primary	BPB	Primary	BPB
Knowledge QA Tasks						
TriviaQA	27.9	61.8	2.5	0.5	68.3	85.3
SQuAD	23.8	29.0	7.6	27.8	59.7	61.7
OLMES Gen	23.1	20.6	0.9	2.6	63.3	67.3
ARC Easy	21.0	64.6	5.3	0.8	93.0	93.0
Jeopardy	20.2	22.6	3.5	18.6	82.0	83.0
AutoBench	15.9	31.3	0.2	4.5	89.3	89.3
HellaSwag	11.8	14.9	1.4	1.0	74.3	95.3
DROP	11.5	9.9	59.0	11.3	57.3	58.7
OLMES + Gen	11.2	40.0	2.1	0.4	89.0	89.0
MMLU Pro	11.0	27.6	2.7	1.3	83.0	89.0
MMLU	9.8	35.9	4.3	0.4	89.0	92.0
ARC Challenge	6.6	44.8	9.7	2.1	83.3	95.0
CommonsenseQA	5.5	41.9	3.6	5.9	68.7	65.7
SocialIQA	5.5	48.0	0.4	1.9	55.0	80.0
OLMES Core 9	5.4	73.2	3.7	0.2	73.3	79.3
WinoGrande	4.6	3.6	10.3	0.9	49.7	75.0
PIQA	4.2	8.8	0.5	1.3	73.3	72.7
BBH	3.6	2.5	67.1	12.9	64.7	55.0
MedMCQA	3.5	29.5	8.8	4.6	60.3	86.7
AGI Eval	2.5	19.5	13.7	3.4	58.7	88.0
OpenBookQA	2.1	24.2	7.7	3.3	65.7	82.7
BoolQ	1.5	64.8	5.1	6.6	47.7	62.3
Knowledge 19-Task Avg.	13.7	44.3	0.8	1.0	79.0	80.0
Math Tasks						
Minerva MATH	1.9	88.6	11.9	1.9	51.0	90.0
GSM+	1.8	7.3	20.0	4.8	59.7	79.0
GSM Symb.	1.3	6.5	83.0	5.1	51.0	78.3
GSM8K	1.2	7.0	38.6	5.9	46.0	76.7
Math 6-Task Avg.	1.8	22.6	46.0	5.0	42.3	88.3
Code Tasks						
HumanEval	6.1	25.1	9.2	7.9	74.3	95.7
HumanEval+	5.5	27.4	29.7	7.1	66.0	96.3
MBPP	2.0	41.8	23.6	1.0	68.3	95.3
MBPP+	1.7	30.8	39.5	8.9	62.7	93.0
GSM Symb. P1	1.6	6.6	538.6	5.2	41.3	81.3
Minerva MATH 500	1.4	90.5	52.5	0.9	50.7	90.3
GSM Symb. P2	1.0	7.0	74.8	5.1	40.3	79.7
Code 4-Task Avg.	5.5	42.0	29.5	9.7	80.3	96.7
All 30-Task Avg.	10.0	31.5	2.3	0.4	77.0	83.7

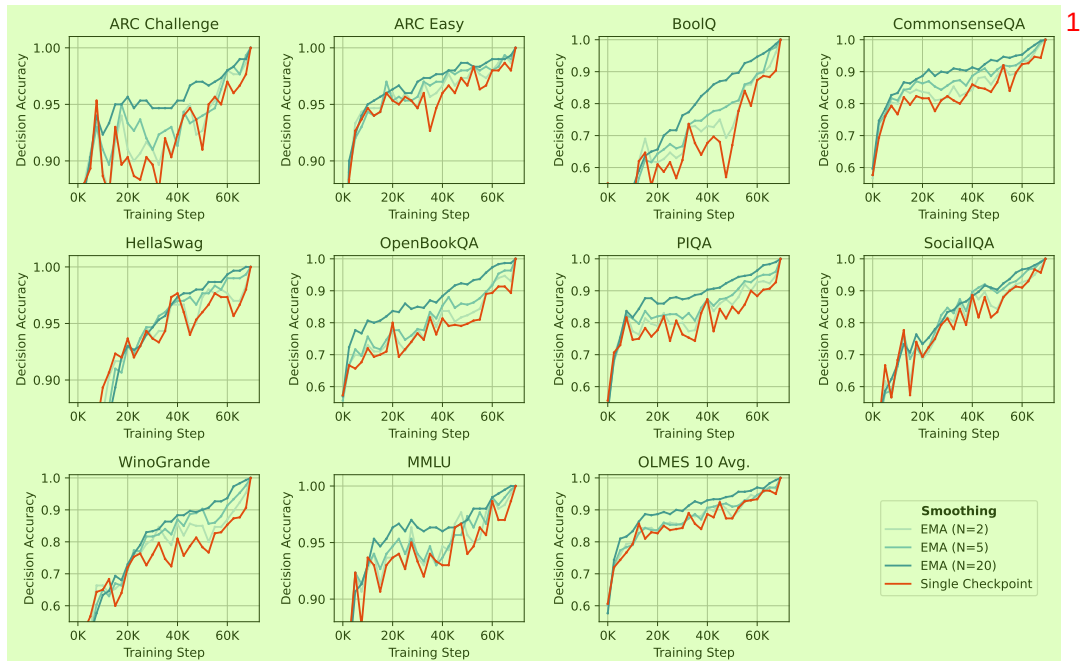


Figure 18: When stopping a training run early, averaging the checkpoint-to-checkpoint noise improves the decision accuracy between an intermediate and the final training step. Shown are decision accuracy from early-stopping for the core OLMES tasks by using both a single checkpoint and the exponential moving average (EMA) 1 2

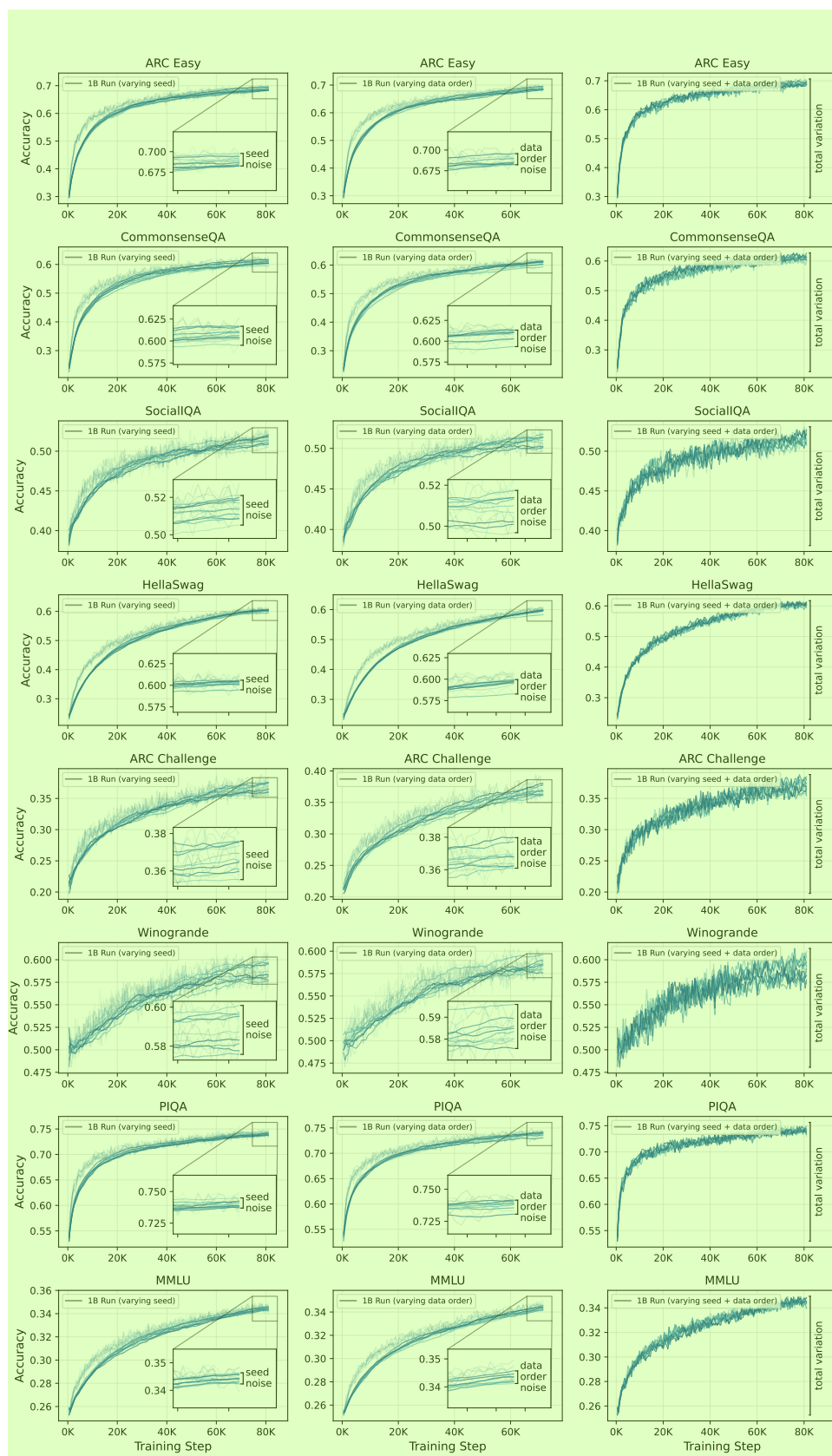


Figure 19: Visualization for the seed noise, data order noise and total variation for all OLMES tasks.

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